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Artificial intelligence for solar water heating systems: A review of global research trends, advances, and future perspectives

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ABSTRACT

Solar water heating systems (SWHS) are pivotal in reducing fossil fuel dependence and promoting sustainable thermal energy solutions. However, the performance of SWHS remains limited by design inefficiencies, dynamic environmental conditions, and limited predictive and control capabilities. This study examined the role of artificial intelligence (AI) in improving the performance, optimisation, and sustainability of SWHS. The study adopted a dual-methodological approach combining quantitative bibliometric analysis and qualitative systematic review. The bibliometric analysis, based on 245 Scopus-indexed documents (2000–2024), was conducted using Bibliometrix (R) and VOSviewer to identify global research trends, collaborations, and thematic clusters. Complementarily, the systematic review provided in-depth qualitative insights into AI applications, including predictive modelling, optimisation algorithms, intelligent control systems, and hybrid designs. The findings show exponential growth in AI-related SWHS research since 2019, mainly driven by contributions from China, India, and the United States. Machine learning, artificial neural networks, and multi-objective optimisation dominate the research domain, revealing significant improvements in thermal efficiency, exergy performance, and cost-effectiveness. Nevertheless, regional disparities persist, with limited contributions from Africa and Latin America despite their high solar potential. The study highlights that integrating AI with SWHS enables data-driven optimisation, predictive maintenance, and adaptive control, leading to smarter, more sustainable, and cost-efficient systems. These insights provide a quantitative and qualitative foundation for guiding future research, policy formulation, and technology development toward intelligent and equitable solar thermal energy solutions.

Introduction

The global energy sector is undergoing a keen transformation due to challenges such as increased energy demand [1], climate change mitigation [2], and the urgent need for sustainable technological solutions [3]. According to the United States Energy Information Administration [4], global energy consumption is projected to grow by nearly 50 % between 2020 and 2050, due to strong economic growth in developing Asian economies. As global energy demand continues to surge, one promising solution is the widespread adoption of solar water heating systems (SWHS). SWHS uses the sun's energy to warm water, reducing the need for energy-intensive traditional water heating methods. Therefore, transitioning to SWHS is vital to reduce carbon emissions [5], decrease fossil fuel dependence [6], and provide cost-effective thermal energy solutions across residential, commercial, and industrial sectors [7].

Global solar water heating capacity peaked at 560 GWth in 2023, an increase of 18 GWth compared to the previous year. China is the world's biggest SWH market. According to [8], between 2009 and 2022, the capacity of SWHS worldwide has increased steadily as new policies have helped improve sales. The global market for water heaters was valued at \$23.7 billion as of 2023. It is expected to reach \$32.1 billion by 2029, growing at a compound annual growth rate of 5.2 % from 2024 through 2029. This substantial increase suggests the rising demand for modern water heating solutions and the industry's ongoing advancements [9].

SWHS comes in various designs, all incorporating a collector and storage tank to utilise the sun's thermal energy to heat water [10]. These systems can be categorised based on the type of collector and the circulation system used. Closed-loop, or indirect, systems use a non-freezing liquid to transfer heat from the sun to the water in a storage tank [11]. In this system, the sun's thermal energy heats the fluid in the solar collectors, which then passes through a heat exchanger in the

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storage tank, transferring the heat to the water. The non-freezing fluid then cycles back to the collectors, making this system useful in freezing climates. Indirect solar water heating systems often use propylene glycol as the heat transfer fluid to prevent freezing and eliminate scaling issues [12]. Another variation, the indirect solar drain-back system, relies on a heat exchanger to separate the collector loop from the potable water, draining the water back into a storage vessel when the pump is off to avoid freezing [13].

Recent advancements in SWHS have focused on improving efficiency and performance. For instance, large-scale solar thermal plants and district heating systems have developed significantly, contributing to decarbonising the heat sector in neighbourhoods and cities [14]. Innovations such as those developed by Mixergy, which use artificial intelligence (AI) to optimise home-produced solar energy, can potentially reduce energy costs for heating water [15]. Moreover, projects like the solar thermal system at Creighton University, which uses vacuum-sealed glass tubes to convert the sun's energy into heat, highlight the ongoing efforts to transition away from fossil fuels for heating purposes [15].

Currently, AI is deployed as an optimisation tool across various renewable energy applications, including solar photovoltaic (PV) systems [16,17], wind farms [18], solar thermal systems [19], hydropower [20], and green hydrogen production [21]. AI algorithms can be applied in SWHS to optimise thermal performance, predict maintenance needs, and improve energy management. For instance, AI-driven weather forecasting models like GraphCast [22] and Pangu-Weather [23] can enable SWHS to dynamically adapt to real-time variations in solar irradiance, thereby maximising energy capture and improving overall system efficiency. In addition, the GraphCast and Pangu-Weather can predict weather conditions up to 10 days in advance with high precision, outperforming traditional methods [24,25]. This precise forecasting optimises energy capture based on sunlight availability. Likewise, smart controllers integrated into SWHS can learn user behaviour patterns, optimise water use and minimise energy loss for a more reliable, cost-effective, and sustainable solution. Similarly, AI can analyse user behaviour, predict demand, and adjust heating schedules based on hot water usage patterns. This proactive approach ensures hot water availability when needed and reduces energy waste during low-demand periods, leading to efficient energy use, cost savings, and greater sustainability of SWHS [26,27].

Numerous studies have applied AI techniques over traditional methods for predicting SWHS thermal performance and optimising design. For instance, Liu et al. [28] and Kalogirou and Panteliou [29] revealed that artificial neural networks (ANN) effectively predict heat collection rates, heat loss coefficients, and solar energy outputs with high accuracy, outperforming conventional regression approaches. Razavi et al. [30] confirmed ANN's improved prediction accuracy across materials, notably polypropylene and copper, emphasising the material's impact on solar heater efficiency. Correa-Jullian et al. [31] further established that long short-term memory (LSTM) models perform exceptionally well in sequence predictions for SWHS, capturing complex temporal patterns with minimal error. Enhanced methodologies like modified shallow swarm optimisation with deep learning by Deepanraj et al. [32] and attention-based LSTM with decomposed input by Heidari and Khovalyg [33] underline AI's potential in refining predictive accuracy for energy usage under fluctuating conditions. Hybrid models combining ANN and Transient System Simulation (TRNSYS) tool, as shown by Souliotis et al. [34], effectively predict annual solar heater performance.

Similarly, other studies have used the traditional method to review the development, advances, and progress of SWHS. For example, Shamsul Azha et al. [35] reviewed existing methods on thermal performance enhancement for flat plate solar collectors and discussed heat-transfer enhancement using vibration and its potential application for flat plate solar collectors. Vengadesan and Senthil [36] analysed recent developments in the thermo-economic performance of SWHS regarding the design modification of thermo-physical properties of heat transfer fluids, integrated thermal energy storage, and hybrid systems of flat plate solar collectors. Uniyal et al. [37] examined recent advancements in evacuated tube solar water heaters integrated with phase change materials (PCM) and nanofluids. Aggarwal et al. [38] comprehensively reviewed techniques for increasing the efficiency of evacuated tube solar collectors. Nanda et al. [39] highlighted energetic and exergetic approaches and analyses, technological development, and examined the advantages, disadvantages, and developments of SWHS. Al-Mamun et al. [10] comprehensively analysed the state of the art of SWHS for sustainable solar energy utilisation. Singh et al. [40] comprehensively assessed the use of PCM in SWHS. Naveenkumar [41] reviewed advancements and modifications in solar water heaters and collectors, focusing on efficiency enhancement techniques and applying ANN to improve their performance, economic viability, and sustainability.

The contribution of this study lies in its dual-methodology framework, which provides a data-driven analysis of AI integration in SWHS. In contrast with previous qualitative reviews that primarily emphasised technical aspects, performance evaluation, and design optimisation through qualitative methods [10,35–41], this study provides a more systematic and evidence-based perspective. This study combines bibliometric analysis with a qualitative systematic review to provide a holistic and quantitative assessment of research trends. The framework goes beyond existing efforts by systematically mapping global research patterns, identifying collaborative networks, pinpointing influential research clusters, and tracking technological advances from 2000 to 2024. This methodological innovation provides evidence-based recommendations and quantitative insights into the research dynamics of AI-optimised SWHS, addressing a critical knowledge gap in understanding the evolution and future trajectory of intelligent optimisation techniques in sustainable solar thermal applications.

The structure of the remaining sections in this paper is as follows: **Section 2** provides a concise overview of the solar water heating systems; **Section 3** describes the methodology adopted in the study; **Section 4** presents the results of the bibliometric analysis along with a discussion of their implications; **Section 5** highlights the insights from the qualitative systematic review. **Section 6** summarises the findings and proposes directions for future research, and **Section 7** outlines the conclusions drawn from the study.

Brief overview of solar water heating systems

SWHS, which use solar energy to heat water, are classified into two main types: active and passive systems (Fig. 1). Active systems rely on pumps and controllers to circulate water or heat-transfer fluid through the solar collectors [42]. In direct active systems, water is pumped directly through collectors and stored, making them suitable for climates where freezing is not a concern [43]. In indirect active systems, a non-freezing fluid such as glycol circulates through the collectors, transferring heat to water via a heat exchanger, which protects against freezing in colder climates [44].

Passive SWHS, on the other hand, do not use electrical components and rely instead on natural convection [10]. Thermosiphon systems place the storage tank above the collector so that heated water rises naturally into the tank as cooler water flows down [46]. Integral collector-storage systems, also known as batch heaters, combine the collector and storage unit, preheating water before it enters a conventional heater [47]. Although active systems generally perform better in variable climates, passive systems are simpler, more durable, and cost-effective, making them ideal for stable, warm regions.

There are several types of solar collectors, each with unique efficiency and applicability. The two main solar thermal collectors are flat plate and evacuated tube collectors [42,48]. Flat plate collectors are the most common and consist of a flat, rectangular box with a transparent cover and a dark absorber plate (Fig. 2a). They are efficient and cost-effective, making them suitable for residential water heating [49]. On the other hand, evacuated tube collectors consist of a series of parallel

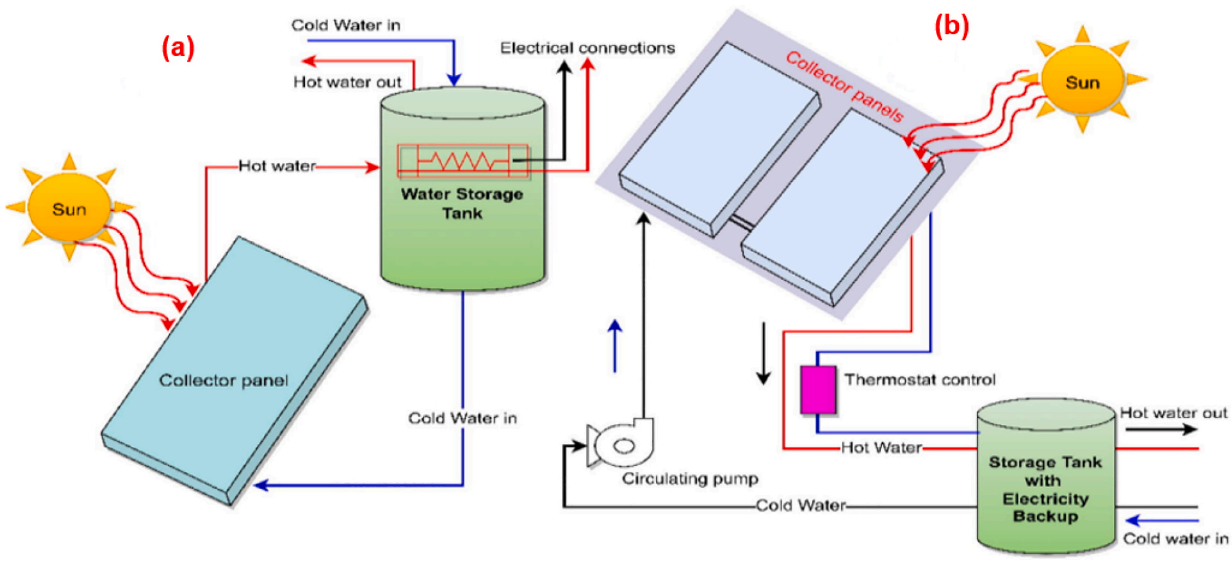
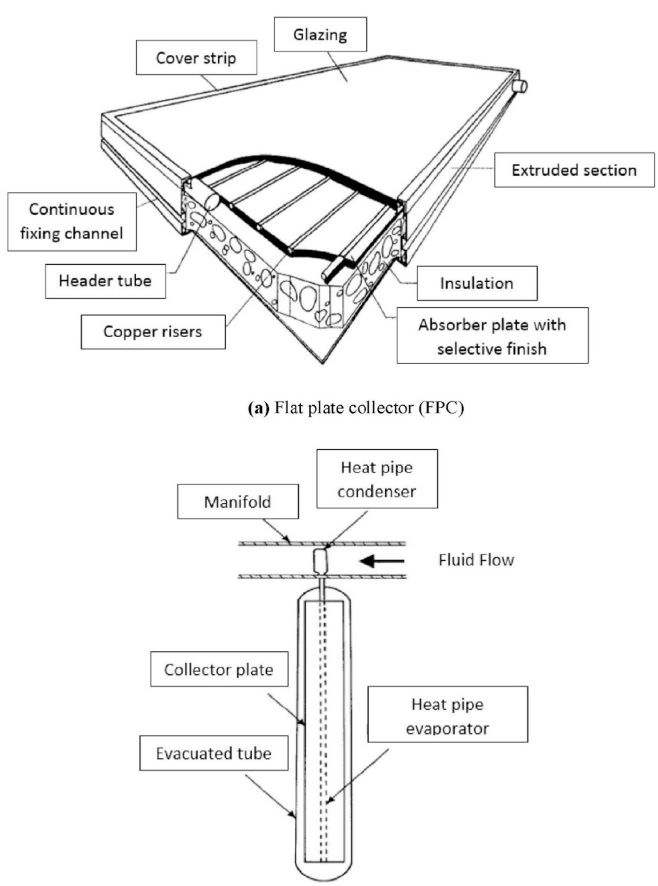


Fig. 1. Schematic of (a) Passive SWHS; (b) Active SWHs [45] (Reproduced with permission from Elsevier, License number: 5907510436562).



(a) Flat plate collector (FPC)

(b) Evacuated tube solar collector (ETC)

Fig. 2. (a) Flat plate collector; (b) Evacuated tube collector [42] (Reproduced with permission from Elsevier, License number: 5907510770775).

glass tubes, each containing an absorber tube covered with a selective coating, as shown in Fig. 2b. Due to their heat retention capabilities, these collectors are highly efficient, especially in colder climates [50].

Another type of collector is the concentrating collector, which uses mirrors or lenses to focus sunlight onto a small, high-efficiency absorber area [42]. This type of collector is often used for larger applications such as industrial processes and electricity production in solar thermal power plants. In addition to these types, solar heat pipes are also used in some systems. Heat pipes efficiently transfer heat from the absorber to the storage system and are commonly used in evacuated tube systems [51]. The collector's choice depends on the specific application, climate, and cost considerations.

Methodology

The study methodology follows a comprehensive and systematic approach to achieve the study goal. Fig. 3 **(a)** displays the data collection process. The study strictly adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines to ensure transparency and reproducibility for the data collection process [52–54]. This study utilised the Scopus database as the primary source for data extraction, chosen for its distinctive advantages over other databases such as Web of Science and Google Scholar. The Scopus database was selected due to its broader coverage of peer-reviewed literature in engineering and technology, more consistent citation tracking, comprehensive metadata availability, and quality control mechanisms for indexed publications [55,56]. In addition, Scopus provides more data export and analysis tools, making it suitable for bibliometric studies [57].

The search strategy was designed using Boolean operators to combine two main concept clusters artificial intelligence-related terms: ("fuzzy logic" OR "artificial neural networks" OR "genetic algorithms" OR "artificial intelligence" OR "machine learning" OR "deep learning" OR "natural language processing" OR "reinforcement learning" OR "support vector machines" OR "Bayesian networks" OR "decision trees" OR "evolutionary algorithms" OR "k-Means clustering" OR "principal component analysis" OR "random forests" OR "gradient boosting machines" OR "deep reinforcement learning" OR "genetic programming" OR "automated machine learning" OR "self-organizing maps" OR "extreme learning machines" OR "long short-term memory" OR "convolutional neural networks" OR "particle swarm optimisation" OR "ant colony optimisation" OR "Markov decision processes" OR "Q-learning") and solar water heating system-related terms: ("solar hot water systems" OR "solar hot water system" OR "solar water heaters" OR "solar water heating" OR "solar water heater" OR "solar water collector" OR "solar water collectors" OR "solar thermal water heating" OR "solar thermal collectors" OR "solar thermal collector" OR "solar water heating systems" OR "solar thermal energy for water heating" OR "solar thermal

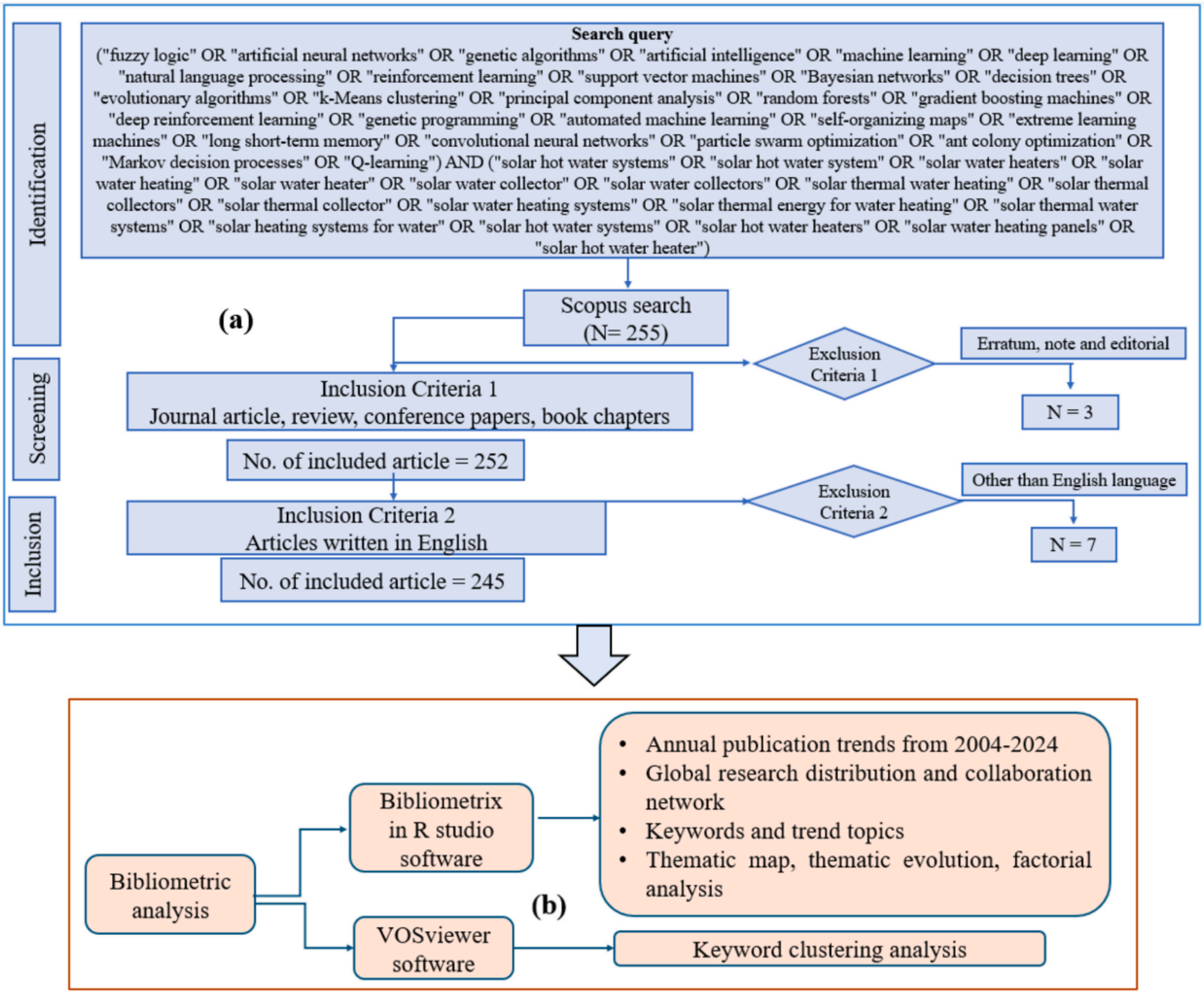


Fig. 3. Study methodology adopted: (a) PRISMA approach adopted for data collection for quantitative and qualitative analysis, and (b) bibliometric analysis framework. Adapted and modified from [58,59].

water systems" OR "solar heating systems for water" OR "solar hot water systems" OR "solar hot water heaters" OR "solar water heating panels" OR "solar hot water heater").

The search was conducted on October 21, 2024, covering publications from 2000 to 2024, yielding an initial dataset of 255 documents, as shown in Fig. 3 **(a)**. The inclusion criteria included research articles, conference papers, review papers, and book chapters. In contrast, the exclusion criteria comprise notes, errata, and editorials. This initial filtering process reduced the dataset to 252 documents. After applying a language filter to include only English-language publications, the dataset was refined to 245 documents. Afterwards, data cleaning was conducted to ensure the dataset's accuracy, consistency, and reliability. Duplicate records were identified and removed using document titles, author names, publication years, and DOIs as reference points. Inconsistent entries, such as variations in author names, keywords, and source titles, were standardised to maintain uniformity across the dataset. Finally, the cleaned dataset was verified in Bibliometrix (R) and VOSviewer by cross-checking to confirm that all retained records were consistent and correctly formatted for subsequent analysis.

Fig. 3(b) displays the bibliometric analysis process using the Bibliometrix R-package (R-version 4.3.2) and VOSviewer (version 1.6.20) software. Bibliometrix facilitates comprehensive bibliometric analysis, enabling data-driven insights, network mapping, and visualisation of literature trends [60–64]. On the other hand, VOSviewer specialises in constructing and visualising bibliometric networks, highlighting relationships such as co-authorship, co-citation, and keyword co-occurrence [65,66]. The Bibliometrix R-package was used to examine the following: (i) annual publication trends; (ii) geographical analysis of research production and international collaborations; (iii) keyword analysis; (iv) thematic mapping and evolution; and factorial analysis to identify underlying research patterns. In contrast, VOSviewer was utilised to perform keyword co-occurrence analysis, clustering the keywords into individual thematic groups.

This study performed a qualitative systematic review to complement the quantitative insights obtained from the bibliometric analysis. The systematic review applied clear inclusion and exclusion criteria to ensure the relevance and quality of the selected studies. The inclusion criteria prioritised peer-reviewed publications focusing on AI applications in SWHS, specifically those addressing predictive modelling, optimisation, control, or system design. The exclusion criteria eliminated studies without explicit AI integration or that did not involve thermal water heating. These criteria ensured that the final dataset comprised the most relevant, high-quality, reproducible studies aligned with the review's objectives. It is worth mentioning that the systematic review aimed to extract in-depth information on technological advancements, methodological approaches, and emerging research directions within the domain.

Results and discussion

Annual trend and temporal distribution of publications

Fig. 4 displays the annual publications trend from 2000 to 2024. The figure shows a clear upward trajectory, with notable patterns and growth over these 24 years. The results show just 3 publications in 2000, followed by intermittent activity in the early 2000 s, as evidenced by years with zero publications (2001 and 2003) and low numbers (1–4 papers) until 2007. A noticeable increase occurred in 2008 with 10 publications, indicating the beginning of increased research interest in this field. The linear regression equation $y = 1.0554x - 2113.6$, with an R^2 value of 0.776, highlights this growth pattern. The positive slope of 1.0554 indicates that, on average, there has been an increase of about one publication per year. At the same time, the R^2 value of 0.776 suggests a strong fit of the trend line to the data, explaining about 77.6 % of the variation in publication numbers over time.

The growth became more evident after 2014, with publications exceeding 10 papers yearly. A significant surge is observed from 2019 onwards, with the number of publications reaching 25 in 2019, followed by sustained high levels of output (17–21 papers annually) between 2020 and 2023, culminating in a peak of 34 publications in 2024. This recent research output suggests a keen interest in AI's potential to optimise SWH systems, coinciding with broader advances in AI technology and increased focus on renewable energy solutions.

It is worth noting that the results show three distinct phases: an initial period of low activity (2000–2007), a transition period of moderate but irregular growth (2008–2018), and a recent period of sustained high output (2019–2024). The positive trend, especially in recent years, implies developing research interest, potentially driven by improved AI capabilities, greater awareness of climate change issues, and the practical benefits of integrating AI into solar thermal systems. The R^2 value suggests that this growth trend will continue, indicating sustained research interest in this field.

Geographical distribution and international research collaboration network

The geographical contribution of research publications is displayed in Fig. 5. It can be observed that China is the leading contributor with 151 publications, representing roughly 22 % of the total output, followed by India with 105 publications (16 %) and the United States with 65 publications (10 %). These three countries account for nearly half (48 %) of all research activity in this domain, indicating their substantial

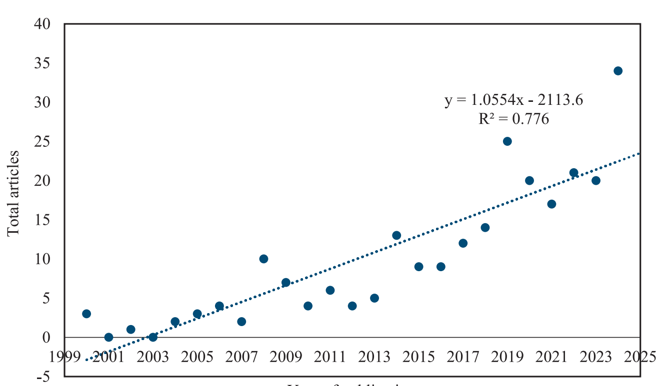


Fig. 4. Annual distribution of article production on AI applications in SWHS research.

investments in renewable energy infrastructure and AI research capabilities. Iran's contribution of 56 publications (8.3 %) positions it as a significant player in the Middle East and North Africa (MENA) region. In addition, contributions from European research countries such as Italy (36 publications, 5.3 %), Spain (34 publications, 5.0 %), and Germany (14 publications, 2.1 %) show an average contribution within this interdisciplinary field.

The research contribution from developing nations shows an interesting pattern, with Brazil and Mexico each contributing 21 publications (3.1 % each), Indonesia and Saudi Arabia each with 19 publications (2.8 %), and Egypt with 18 publications (2.7 %). However, the figure shows a troubling pattern when examining the contribution of African countries. The entire African continent contributes only 36 publications combined (about 5.3 % of global output), with Egypt leading at 18 publications, South Africa and Cameroon each with 4 publications (0.6 %), Algeria with 5 publications (0.7 %), Morocco with 3 publications (0.4 %), and Ethiopia and Tunisia each with a single publication (0.1 %). This marginal contribution is alarming due to Africa's enormous solar energy potential, with the continent recording more hours of bright sunshine annually and the pressing need for sustainable, affordable water heating solutions across African communities.

Africa faces significant challenges in energy access, with over 600 million people lacking electricity [67] and many more relying on inefficient, polluting energy sources for basic needs, including water heating. The application of AI to optimise solar water heating systems could transform energy access, reduce carbon emissions, and improve the quality of life across the continent. However, with African nations contributing less than 6 % of research in this field, there is a critical risk that AI-driven solar water heating solutions will be designed without consideration for African contexts, climatic conditions, economic constraints, and specific user requirements. This could lead to technology transfer failures, suboptimal system performance, and missed opportunities for indigenous innovation. Therefore, building African research capacity in AI-enhanced SWHS is not merely an academic imperative but an existential necessity for sustainable development, energy independence, and climate change mitigation across the continent.

Furthermore, China, India, and the USA emerging as the major contributors to this research domain could be attributed to funding and access to experimental and simulation infrastructure. In contrast, limited research capacity, data availability, and high equipment costs could hinder similar progress in regions such as Africa and parts of Latin America. This disparity highlights a pressing need for targeted funding, regional test facilities, and open-data collaborations to democratise AI-based SWHS research globally.

The collaboration map in Fig. 6 shows the international research partnerships and the complex network of global scientific cooperation. The map depicts varying intensities of collaboration through shades of blue, with darker blue indicating stronger research partnerships. China and the United States are the research central hubs in this collaborative network, as shown by their darker blue colouring and the connecting lines, suggesting substantial bilateral research cooperation. The map demonstrates research activity across North America, Europe, and Asia, with notable interconnections between these regions as indicated by the connecting lines. European nations maintain active collaboration networks, as suggested by the consistent light blue shading, indicating moderate levels of international cooperation. The Asia-Pacific region, mainly centred around China, shows strong research partnerships extending to Western and neighbouring Asian countries. The collaboration patterns suggest a mostly Northern Hemisphere-centric research community. However, there is evidence of participation from countries in the Southern Hemisphere, as indicated by the lighter blue shading in regions such as Australia and parts of South America. The connecting lines on the map primarily emanate from major research centres, suggesting that smaller nations often collaborate with these larger research hubs rather than directly with each other. This pattern suggests a typical flow of knowledge and resources in international research, where

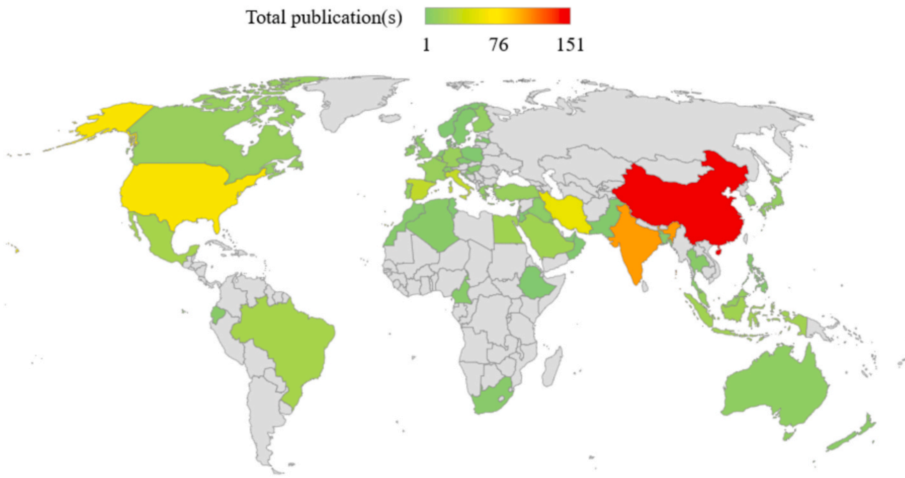


Fig. 5. Country-wise contribution to AI applications in SWHS research.

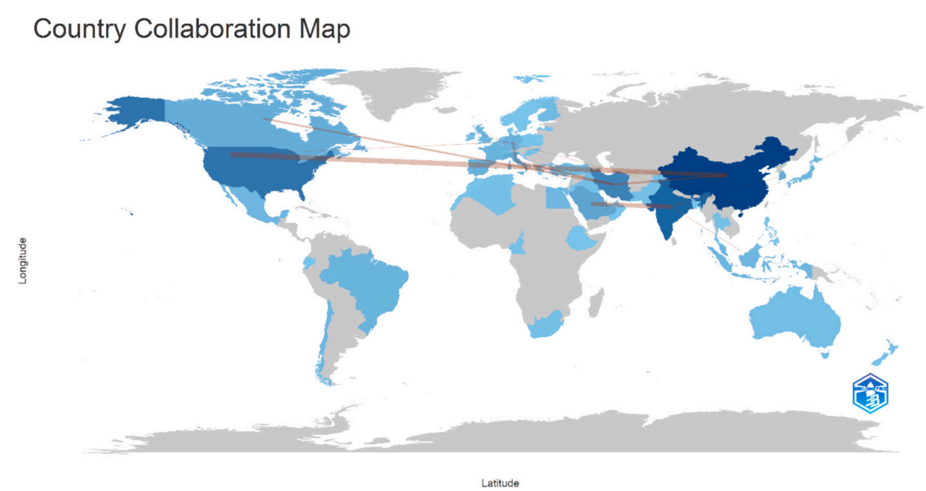


Fig. 6. Scientific collaboration map of AI applications in SWHS research.

established research institutions often serve as bridges for global collaboration.

In some cases, the intensity of collaboration appears to be influenced by geographical proximity. However, long-distance partnerships, specifically those across continents, are also evident through the connecting lines crossing oceans. The map shows relatively less intensive

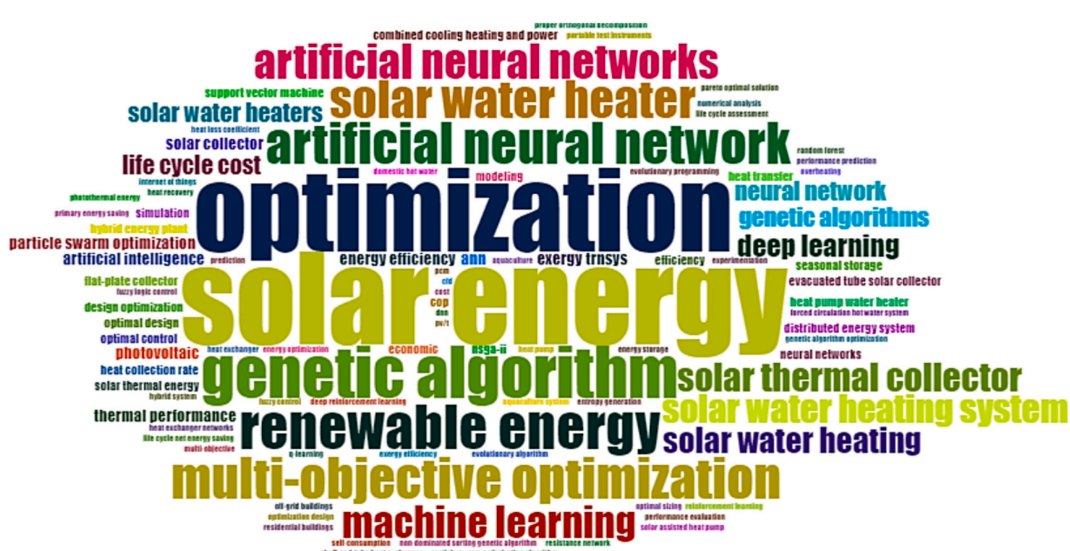


Fig. 7. Word cloud of keywords.

collaboration in Africa and South America, as shown by the lighter shading or grey areas, suggesting potential opportunities for expanding research partnerships in these regions. The overall pattern of collaboration demonstrates the global nature of research in this field, highlighting how countries are working together to advance the integration of AI in SWHS, though with varying degrees of involvement and partnership intensity across different geographical regions. This map effectively captures the international dimension of scientific cooperation in sustainable energy research, showing the strengths of existing research networks and areas where collaborative relationships could be further developed.

Word cloud visualisation of dominant keywords

Fig. 7 presents the word cloud of the top 50 author keywords. It can be seen that the most dominant terms in the figure are "optimisation," "solar energy," and "artificial neural networks." This highlights the fundamental role of AI-driven optimisation approaches in advancing solar thermal technologies. The presence of various AI methodologies, including genetic algorithms, machine learning, deep learning, and particle swarm optimisation, demonstrates the various computational tools employed to enhance system performance and efficiency. These advanced algorithms are applied to multiple aspects of SHWS, from design optimisation to operational control strategies.

The emphasis on "multi-objective optimisation" indicates that researchers are addressing complex problems requiring the consideration of multiple competing factors, such as system efficiency, cost-effectiveness, and environmental impact. The presence of specific solar technology terms like "solar thermal collector," "evacuated tube solar collector," and "flat plate collector" emphasises the components optimised through AI applications. In contrast, the "solar water heating system" represents the integrated system-level approach. The inclusion of "life cycle cost" and "economic" among the keywords suggests the crucial consideration of financial viability and long-term sustainability in system design and implementation. Performance-related terms such as "thermal performance," "energy efficiency," and "heat collection rate" indicate the key metrics targeted for improvement through AI optimisation techniques. The appearance of "exergy analysis" suggests that researchers conduct detailed thermodynamic evaluations to assess and optimise system performance from an energy quality perspective. The presence of "simulation" points to the vital role of computational modelling in developing and testing AI-driven solutions before real-world implementation.

Combining "renewable energy" with various AI techniques indicates a broader focus on sustainable energy solutions, with SWHS indicating a specific application area within this domain. The terms "support vector machine" and "neural network" suggest that researchers leverage traditional and modern machine learning approaches to develop intelligent control systems and prediction models. Terms like "distributed energy systems" and "combined cooling, heating, and power" indicate that the research extends beyond standalone solar water heaters to more complex integrated energy systems. The interconnection of these keywords shows a research ecosystem where AI is systematically applied to enhance every aspect of SWHS, from component design to system integration and operational optimisation. The diversity of AI techniques demonstrates researchers' comprehensive approach to addressing the challenges in solar thermal energy systems. The integration of AI with solar technology aims to improve system efficiency, reduce costs, enhance reliability, and contribute to the broader goal of sustainable energy development. The research area depicted by these keywords suggests a strong focus on developing intelligent, efficient, and economically viable solar water heating solutions by applying cutting-edge AI methodologies, signifying a vital intersection of renewable energy technology and computational intelligence.

Keyword clustering analysis

Fig. 8 shows the network visualisation of keyword co-occurrence obtained via VOSviewer software. It is worth noting that a minimum threshold of four occurrences was applied, resulting in 26 keywords from a total of 650 that met this criterion. The network visualisation was generated using the association strength normalisation method with default layout parameters (attraction: 2, repulsion: 0). For clustering analysis, a resolution of 1 and a minimum cluster size of 1 were set, with the option to merge small clusters enabled. Each cluster, characterised by its collective set of keywords and thematic focus, is discussed in detail to highlight its significance within the research landscape. In addition, qualitative insights from the relevant literature are integrated to substantiate and enrich the interpretation of each cluster. The following sub-sections present an in-depth discussion of the five identified clusters.

Computational intelligence and optimisation framework

Cluster 1 comprises keywords such as "artificial neural network (ANN)", "machine learning", "multi-objective optimisation", "particle swarm optimisation (PSO)", "renewable energy", "solar energy", and "solar thermal collector". This cluster indicates the foundational computational intelligence methodologies underpinning modern solar water heating research. Artificial neural networks and machine learning are pivotal in predictive modelling and pattern recognition, allowing accurate forecasting of complex, nonlinear interactions in solar thermal systems. Sundar and Chandra Mouli [68] applied a Levenberg-Marquardt artificial neural network to predict heat transfer and friction factors in thermosyphon flat plate collectors using MgO/water nanofluids, achieving correlation coefficients above 0.98, confirming high predictive accuracy between experimental and simulated data. Similarly, Mausam et al. [69] employed an ANN to model the thermal performance of solar flat plate collectors using Cu-MWCNTs hybrid nanofluids, attaining coefficient of determination (R2) values up to 0.9989. Machine learning's versatility extends to ensemble methods, where Said et al. [70] utilised Boosted Regression Tree and XGBoost algorithms to predict Nusselt numbers, friction factors, and efficiencies in hybrid nanofluid-based collectors. The authors obtained near-perfect R^2 values (0.9914–0.9997), indicating their capacity to generalise performance behaviour across diverse flow conditions. Farhadi et al. [71] advanced this domain by integrating analytical modelling and machine learning to predict the performance of heat pipe evacuated tube solar collectors, identifying Extra Trees and LightGBM as best in accuracy with a correlation coefficient of 0.995. Uniyal et al. [72] reinforced this by comparing nine machine learning models to predict U-tube PCM solar collector performance. The ANN and support vector regression (SVR) yielded higher accuracies (R^2 up to 0.9540), highlighting the importance of model selection for system-specific optimisation.

Similarly, including multi-objective optimisation and PSO in this cluster highlights the critical role of metaheuristic optimisation in refining solar collector performance. For example, Rey and Zmeureanu [73] applied micro-time variant multi-objective PSO to a residential solar thermal combisystem, balancing life-cycle cost and energy consumption. The results showed that increasing solar collectors from one to ten decreased life-cycle energy use by 63 % but increased cost by 84 %, emphasising the necessity of trade-off decision-making. Mohan et al. [74] adopted multi-objective PSO to optimise a solar minichannel flat plate collector, identifying optimal flow rates of 0.01–0.015 kg/s and improving efficiency by 10–12 % compared to conventional systems. Likewise, Dong et al. [75] implemented an enhanced PSO algorithm for a hybrid photovoltaic/thermal (PV/T) collector, achieving a 17.93 % reduction in pump energy consumption and improved thermal efficiency by 7.86 %. Summ et al. [76] integrated machine learning metamodels and a genetic algorithm for optimising insulating glass flat-plate collectors. The authors identified argon concentration and collector geometry as key parameters influencing efficiency and cost-

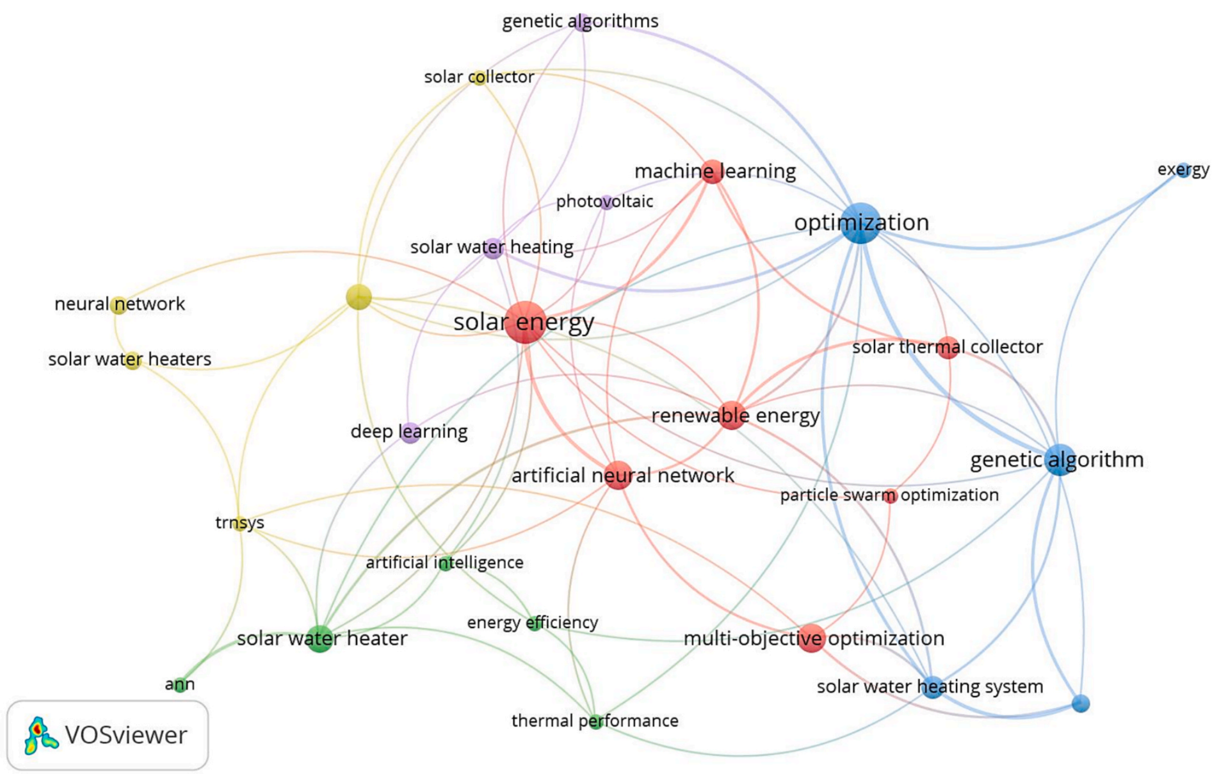


Fig. 8. Network visualisation of keyword co-occurrence relationships.

effectiveness, marking a methodological advancement in multi-objective optimisation frameworks. Mousa et al. [77] applied multi-objective optimisation to determine the optimal technological mix of solar PV and thermal collectors for industrial applications. It was observed that hybrid configurations outperform single-technology systems in energy, cost, and environmental indicators, reinforcing optimisation's strategic role in solar system design. It worth mentioning that these studies emphasize that multi-objective optimisation and PSO offer efficient solutions for balancing conflicting objectives, enabling intelligent and adaptive control strategies in renewable energy systems.

Furthermore, the broader contextual keywords such as renewable energy, solar energy, and solar thermal collectors define the application domain driving this technological transformation. Solar thermal collectors are the principal hardware elements that convert solar radiation into usable heat, with ongoing innovations enhancing operational efficiency. Hajabdollahi [78] optimised an integrated liquid solar thermal collector with a cross-flow heat exchanger, achieving maximum efficiency of 0.6359 by adjusting collector geometry and mass flow rates. Raja Sekhar et al. [79] applied explainable AI to evaluate grooved absorber tubes in solar flat plate collectors. The study identified groove pitch and solar radiation as critical factors that improved efficiency by up to 20 % compared to plain tubes. Skheel et al. [80] combined ANNs and genetic algorithms for geometry optimisation of solar collectors, identifying parabolic-curved designs as globally optimal, improving energy absorption by 15 % compared to conical designs, reinforcing the integration between computational intelligence and design innovation. Sridharan [81] applied the XGBoost algorithm to predict the thermal performance of modified flat-plate water collectors, achieving an accuracy of 99.8 %.

It is worth noting that Cluster 1 shifts from empirical experimentation to intelligent, data-driven solar energy system optimisation. ANNs and machine learning methods allow precise performance predictions; multi-objective optimisation and PSO balance efficiency and cost trade-offs; and renewable energy-focused design innovations extend the applicability of solar thermal collectors. As demonstrated in the studies, integrating these techniques improves energy conversion efficiency and ensures sustainability and adaptability across varying environmental and operational contexts. This convergence of computational intelligence and solar energy research underlines an advancement in renewable energy engineering, propelling the development of self-optimising, predictive, and efficient SWHS suited for global deployment.

Intelligent systems and performance analysis

Cluster 2 contains the following terms: "ANNs," artificial intelligence, "energy efficiency", "solar water heater", and "thermal performance". Cluster 2 applies AI, specifically ANNs, to evaluate and enhance performance characteristics. Due to their universal approximation capabilities, ANNs have become the primary AI technique for modelling solar water heater behaviour. These networks learn from empirical data without requiring explicit mathematical formulations of complex physical processes. Xu et al. [82] demonstrated ANN efficiency in optimising energy parameters for a carbon dioxide (CO_2) air source heat pump water heater. The authors validated that the ANN provided energy predictions for optimisation of the non-dominated sorting genetic algorithm under low ambient temperatures. The final optimised solution achieved CO_2 emissions of 8599.4 kg and a total annual cost of 1626.9 USD/yr. Sadeghi et al. [83] employed multi-layer perceptron and radial basis function techniques for predicting modified evacuated tube solar collector performance. The empirical data-driven approach predicted temperature differences and energetic efficiency with maximum mean relative percentage errors of 0.576 and 0.907, respectively. The results demonstrated that both multilayer perceptron (MLP) and radial basis function (RBF) reliably predict thermal characteristics of solar collectors using nanofluids.

The keyword "energy efficiency" in this cluster occupies a central position, showing the primary objective of AI applications in solar water heating optimisation. Energy efficiency considerations include heat transfer efficiency, insulation performance, optimisation of circulation, and temporal energy supply–demand matching. In view of this, Prakasam et al. [84] investigated flat-plate SWHS performance using cerium oxide–water nanofluid through holistic analytical approaches. The study refined the ANN model with a 3–10-2 architecture, which achieved a mean squared error (MSE) of 8.2×10^{-3}. In addition, the system achieved a maximum thermal efficiency of 78.2 % at a 2 L/min flow rate. Heat transfer coefficient enhancements led to 21.5 % thermal efficiency increases compared to plain water systems. Pathak et al. [85] applied machine learning insights to fin-integrated heat pipe evacuated tube solar collector systems with thermal battery backup. The ANN model effectively aligned experimental results with comprehensive thermal performance and sustainability assessment predictions. It was observed that porous aluminium fin-based systems achieved maximum energy enhancement ratios of 24.07 % under full-day operation.

The inclusion of "thermal performance" in Cluster 2 indicates how effectively solar water heaters fulfil intended functions under real-world conditions. This comprises outlet water temperature, response time, thermal storage capacity, and overall system stability characteristics. In view of this, Saravanan et al. [86] evaluated V-trough solar water heater thermal performance using four distinct neural models, including adaptive neuro-fuzzy inference system (ANFIS), generalised linear regression, and support vector machines (SVM). The ANFIS model achieved 0.9990, 0.9966, and 0.9997 correlation coefficients for Nusselt number, friction factor, and efficiency, respectively. ANFIS outperformed other models with root mean square error (RMSE) values of 0.0805, 0.0004, and 0.4534, demonstrating the highest prediction accuracy. Chidiac et al. [87] developed adaptive overheating covers for solar water heaters using an ANN to optimise control system designs. The system's self-organising map network automatically adjusted solar collector temperatures through fabric screen coverage depending on weather conditions and demand. Amasyali et al. [88] proposed model-free deep reinforcement learning approaches for smart water heater control to reduce electricity consumption and emissions. The study used deep Q-networks algorithm trained reinforcement learning agents, achieving electricity savings ranging from 12 % to 22 %. Emission reduction-focused agents reduced CO_2 emissions by 18 % to 37 % by leveraging variable marginal operating emissions rates. Bouguergour et al. [89] investigated combined solar underfloor heating and domestic hot water systems using machine learning-based identification approaches. The PieceWise Affine Auto-Regressive eXogenous model identified four distinct operational states correlating solar radiation with thermal responses.

The specific mention of "solar water heater" as a distinct keyword emphasises this research cluster's practical, application-oriented nature. Unlike more theoretical computational methods, this grouping demonstrates empirical studies where AI techniques directly solve concrete engineering challenges. The integration of these keywords suggests research focused on developing intelligent diagnostic tools, predictive maintenance systems, and performance optimisation algorithms. The significance of this cluster lies in its potential to bridge gaps between theoretical AI development and practical renewable energy applications. These methodological advances contribute to more reliable, efficient, and economically viable solar thermal systems for residential and commercial deployments.

Economic optimisation and evolutionary algorithms

Cluster 3 contains the following terms: "exergy", "genetic algorithm", "life cycle cost", "optimisation", and "solar heating system". Cluster 3 suggests a research focus emphasising economic viability and thermodynamic excellence through evolutionary computational approaches. The presence of "exergy" as a keyword distinguishes this cluster with its focus on second-law thermodynamic analysis. Exergy analysis assesses the quantity and quality of energy transformations within SWHS. This approach provides insights into irreversibilities and thermodynamic inefficiencies that conventional energy analysis might overlook. Ponnusamy et al. [90] conducted exergetic characterisation of flat plate solar collectors employing genetic algorithms to optimise exergetic efficiency. The study controlled the loss coefficient using four decision variables: heat flux rate and collector plate temperature. Exergy correlations were developed across comprehensive operating ranges comprising mass flow rates, inlet temperatures, collector plate areas, and incident solar energy. The results revealed that exergy efficiency increased by 0.01 % to 0.08 % over 24-hour periods. Through experimental testing, Wenceslas and Ghislain [91] validated exergy optimisation of flat-plate solar collectors in thermosyphon solar water heaters. The genetic algorithm-based computational code identified optimal design parameter combinations that maximised exergy efficiency and achieved high performance with reduced collector surface area.

Genetic algorithm serves as the primary computational methodology in this cluster, representing an evolutionary optimisation approach inspired by natural selection processes. Genetic algorithms are well-suited for optimising SWHS because they handle discrete and continuous variables simultaneously. These algorithms accommodate multiple constraints and avoid premature convergence to suboptimal solutions through population-based search strategies. Ko [92] demonstrated a genetic algorithm to optimise indirect forced circulation SWHS based on life cycle cost. The optimisation method incorporated installation and operation-related design variables alongside capacity-related parameters. The authors found that variable inclusion reduced life cycle cost by about 3.2 % and 6.1 %, compared to methods considering only capacity-related variables. Ko [93] further employed genetic algorithms to optimise SWHS configuration and sizing for an office building in South Korea. The study integrated three constraint conditions: energy balance, solar fraction, and available space for solar collector installation. It was observed that low solar fractions do not always present life cycle cost decreases. Trade-offs between equipment and energy cost resulted in optimal designs yielding minimum life cycle cost values.

The inclusion of "life cycle cost" introduces a critical economic dimension to the optimisation framework. Life cycle cost analysis comprehensively comprises initial capital investment, operational expenses, maintenance costs, system lifespan, and salvage value [94]. When integrated with AI-driven optimisation, life cycle cost considerations can identify economically optimal system configurations over entire lifetimes. Silva et al. [95] performed parametric optimisation of stamped longitudinal vortex generators inside circular tubes using genetic algorithms. Their Non-dominated Sorting Genetic Algorithm II (NSGA-II) optimisation method and finite volume numerical simulation enhanced heat transfer for solar water heaters at low Reynolds numbers. Optimal longitudinal vortex generator shapes improved heat transfer by over 50 % across all evaluated Reynolds numbers. The highest augmentation of 62 % occurred at Reynolds number 900, with the best thermo-hydraulic efficiency at Reynolds number 600.

The broader term "optimisation" as a standalone keyword indicates the central methodological theme unifying this cluster. Ahmadi et al. [96] employed a multi-objective NSGA-II for optimising vacuum membrane distillation and compact SWHS. The optimisation minimised water production cost and maximised water production through simultaneous parameter adjustment. The NSGA II algorithm identified optimal system parameters, including 20 membrane rows per collector and 334.4 kg feed mass. Optimised conditions achieved a daily production capacity of 92.5 kg water with a 12.2 USD/m^3 production cost. Ramesh et al. [97] utilised Red-Billed Blue Magpie optimisation alongside NSGA for optimising heat transfer fluid flow rates and PCM melting points. The comparative analysis yields 90 % efficiency with error rates of only 0.035 %. Summ et al. [76] optimised the multi-objective design of insulating glass flat-plate collectors using elitist genetic algorithms. The optimisation identified argon concentration, collector length, and width as critical parameters influencing efficiency and cost-effectiveness. The results revealed 7.7 % point efficiency increases, 19.4 % material cost reductions, and 14.5 % weight decreases compared to market-available collectors. Skheel et al. [80] synergistically deployed ANN and genetic algorithms for optimising multi-objective solar collector geometric parameters. The optimisation convergence revealed that parabolically curved collectors constituted the global optimum, yielding at least 15 % efficiency gains over conical designs.

The implications of this cluster are highly relevant to policymakers, investors, and system designers who require robust economic and thermodynamic justification for solar thermal projects. Researchers have developed decision-support tools that enable more rational and evidence-based investments in solar thermal technologies by integrating genetic algorithms with exergy analysis and lifecycle cost assessment. This cluster highlights ongoing research to determine the system configurations that yield the highest long-term economic returns. It further explores how thermodynamic efficiency can be balanced with financial limitations and identifies the design parameters most strongly influencing lifecycle performance.

Neural network architectures and system components

Cluster 4 includes “artificial neural networks”, “neural network”, “solar collector”, “solar water heaters”, and “TRNSYS” (Transient System Simulation Tool). Cluster 4 focuses uniquely on neural network methodologies and their integration with solar water heating analysis simulation environments. The presence of both “artificial neural networks” and “neural network” as separate keywords suggests variations in terminology across different research communities or emphasis on different neural network architectures. This redundancy also indicates the absolute centrality of neural network approaches in this research subdomain, potentially comprising feedforward networks, recurrent neural networks, radial basis function networks, and deep learning architectures. TRNSYS is a critical software platform widely employed in solar thermal research for dynamic system simulation [98–100]. The appearance of TRNSYS in this cluster suggests research integrating AI methodologies with established simulation frameworks, creating hybrid modelling approaches that combine physics-based simulation with data-driven learning. Souliotis et al. [34] demonstrated this integration by combining ANN with TRNSYS to model an integrated collector storage prototype. The ANN was trained using experimental data from outdoor tests and implemented through the TRNSYS Excel interface for annual performance prediction. This approach effectively merged TRNSYS radiation processing capabilities with the advantages of a neural network’s black box modelling.

The terms “solar collector” and “solar water heaters” denote the specific hardware components and integrated systems under investigation. Solar collectors show complex thermal behaviour influenced by solar geometry, optical properties, thermal losses, and fluid dynamics. Neural networks can model these intricate relationships, potentially replacing computationally expensive simulation runs with rapid surrogate models. Wongsuwan and Kumar [101] compared TRNSYS and ANN methods for predicting forced circulation solar water heater performance under tropical conditions. The study evaluated collector outlet temperature and useful energy predictions across clear, partly cloudy, and cloudy conditions. Both methods yielded satisfactory prediction accuracy for hourly and daily estimations, validating neural networks as viable alternatives to traditional simulation approaches. He et al. [102] advanced this field by developing a real-time fault detection system for solar hot water systems. The hierarchical adaptive resonance theory (ART) neural network was trained using TRNSYS-generated data from verified system models. Using only standard operational sensors, the system detected anticipated faults, including pump failures, impeller degradation, and thermosyphon issues.

Likewise, cluster 4 suggests a methodological bridge between component-level analysis and system-level performance evaluation. Neural networks trained on collector performance data can be computationally efficient models within larger system simulations. System-level neural networks can predict integrated performance based on component characteristics and operating conditions. Correa-Jullian et al. [31] explored deep learning techniques, including recurrent neural networks (RNNs) and LSTM architectures for solar thermal system prognosis. The TRNSYS-generated synthetic data allowed training under various meteorological conditions. The LSTM models achieved the best performance with a mean absolute error (MAE) of 0.55 °C and RMSE of 1.27 °C for temperature predictions. Al-Mamun et al. [10] utilised TRNSYS to simulate SWHS coupled with absorption heat transformers for continuous water heating. Their simulations across Mexican cities showed that integrated systems achieved higher solar fractions up to 99.6 % with reduced collector areas. Rehman et al. [103] employed TRNSYS to optimise centralised and semi-decentralised solar district heating systems in Finnish conditions. The multi-objective optimisation revealed that decentralised systems achieved 35 % life cycle cost reductions and maintained 90 % renewable energy fractions. Narayanan et al. [104] developed coupled TRNSYS-MATLAB frameworks for model predictive control of integrated residential renewable energy systems. The PSO algorithm performed better for complex time-varying system optimisation problems.

The significance of this cluster lies in its contribution to advancing practical engineering design tools. By integrating the flexibility of neural networks with the robust simulation capabilities of TRNSYS, researchers have developed hybrid models that can accurately predict system performance and maintain relatively low computational demands. These tools allow rapid design iterations, sensitivity analyses, and optimisation studies that would otherwise be costly and time-consuming if conducted solely through experimental or full-scale simulations. The methodological innovations highlighted in this cluster have the potential to accelerate advancements in solar thermal technologies by effectively bridging empirical data with theoretical modelling. This multi-scale modelling approach is crucial for the comprehensive analysis and optimisation of SWHS.

Advanced artificial intelligence methodologies and emerging technologies

Cluster 5 comprises “deep learning”, “genetic algorithms”, “photovoltaic”, and “solar water heating”. This cluster suggests the research frontier, incorporating advanced artificial intelligence methodologies and broader solar energy technologies. Deep learning distinguishes this cluster as incorporating state-of-the-art AI techniques characterised by multiple hidden layers and sophisticated architectures. These architectures include convolutional neural networks, recurrent neural networks, and transformer models capable of hierarchical feature learning. Deep learning’s inclusion suggests research addressing more complex challenges that benefit from representation learning capabilities. Correa-Jullian et al. [31] exemplified this advancement by exploring deep learning techniques for solar thermal system prognosis. The study compared ANN, recurrent neural networks (RNN), and LSTM architectures for performance prediction under varying meteorological conditions. The LSTM models yielded better temporal behaviour replication with a MAE of 0.55 °C and a RMSE of 1.27 °C. These results outperformed naïve predictors and traditional regression models, including Bayesian ridge, Gaussian process, and linear regression. Ahmad and Alsalmah [105] further advanced deep learning applications by training RNN models on numerical simulation data. The RNN model predicted temperature distribution across solar-irradiated cylindrical heat exchangers with accuracy. The model achieved a mean squared error (MSE) below 10^{-4} and a relative percentage error of about 2.5 %.

Likewise, the reappearance of genetic algorithms in this cluster suggests its versatility and continued relevance alongside newer AI methodologies. The co-occurrence of genetic algorithms with deep learning potentially indicates hybrid approaches where evolutionary algorithms optimise deep learning architectures. Bahlawan et al. [106] demonstrated the use of a genetic algorithm in optimising hybrid energy plant designs incorporating solar thermal collectors. The optimisation minimised primary energy consumption throughout the plant life cycle by integrating cumulative energy demand assessments. The results showed that life cycle assessment integration achieved about 12 %

higher primary energy savings. As a key term, "photovoltaic" introduces an intriguing cross-technology dimension, suggesting research exploring integrating solar thermal and solar electric technologies. This indicates studies on hybrid photovoltaic-thermal (PV/T) systems that simultaneously generate electricity and useful heat [107]. In view of this, Dimri and Ramousse [108] employed genetic algorithms for thermoeconomic optimisation of solar combined heating and power systems. The study compared four solar configurations, including PV panels with solar-assisted heat pump and PV/T hybrid collectors. Genetic algorithm-based optimisation determined optimal solar panel numbers and electrical and thermal storage capacities for each configuration. Solar-assisted heat pump with PV panels achieved 41 % lower unit product costs than reference systems. Ren et al. [109] proposed a novel planning method for hybrid energy systems integrating PV/T collectors. The modified NSGA with adaptive simulated binary crossover optimised multi-objective models considering economic, energy, and environmental performance. The optimal PV panel capacity reached 465 kW in baseline scenarios but reduced to 357 kW when electrical matching performance was prioritised.

Furthermore, Zeng and Long [110] conducted multi-objective optimisation studies of combined cooling, heating, and power systems integrated with PV/T collectors. The genetic algorithm-based optimisation was applied across five Chinese climate zones. Beijing's cold climate achieved optimal performance with 46.06 % primary energy saving rates and 60.4 % CO_2 emission reduction rates. The implications of Cluster 5 are forward-looking regarding breakthrough potential in complex fault diagnosis and predictive control. Deep learning methodologies provide solutions where traditional approaches have plateaued, mainly in weather pattern recognition for large-scale installations. The inclusion of photovoltaic terminology indicates the broader renewable energy ecosystem within which solar water heating research is situated. This cluster essentially maps the trajectory of future research, indicating that SWHS optimisation will leverage cutting-edge developments. These developments integrate artificial intelligence with comprehensive solar energy challenges, expanding the scope to comprise broader integration objectives.

Conceptual structure (factorial analysis, thematic evolution and mapping)

Fig. 9 displays the factorial analysis. The factorial analysis shows individual clusters and relationships in the research area. The analysis is presented in a two-dimensional space, where different aspects of the research domain are organised into colour-coded clusters.

The upper portion (shown in cyan) focuses on optimisation strategies, highlighting the connection between Pareto optimal solutions, genetic algorithms, and non-dominated sorting, which are crucial for energy-saving applications in lifecycle analysis. The central and largest cluster (shown in purple) demonstrates the core research concentration, comprising several interconnected elements: multi-objective optimisation, artificial neural networks, support vector machines, and optimisation design specifically for SWHS. This cluster also includes life cycle cost considerations and aquaculture system applications, suggesting a broad scope of implementation scenarios. This central cluster indicates that researchers integrate various AI techniques to address multiple aspects of SWHS design and operation.

The lower portion of the analysis (shown in green) focuses on technical performance metrics, including heat collection rate, heat loss coefficient, specific considerations for evacuated tube solar water heaters, and portable test instruments for system evaluation. On the right side (in red), experimental and simulation-based research components are connected through exergy analysis, including flat-plate collector design and water heater implementation. These elements' spatial distribution and clustering suggest that research in this field is well-structured across multiple dimensions, from theoretical AI implementations to practical engineering applications. The positioning of artificial neural networks and support vector machines near the centre of the purple cluster indicates their fundamental role in advancing this field. At the same time, their connections to optimisation design and multi-objective optimisation show the integrated approach researchers are taking to solve

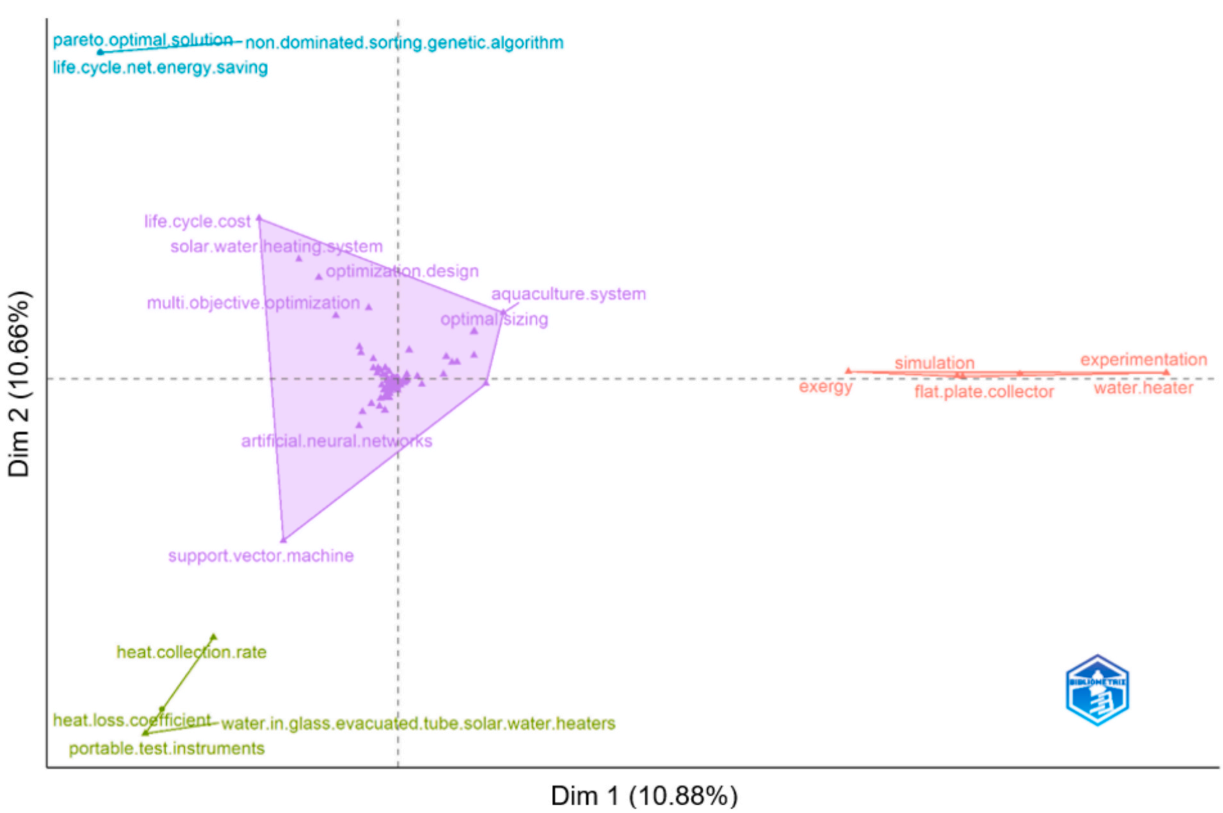


Fig. 9. Factorial mapping of keywords.

complex problems in SWHS. The analysis also demonstrates how life cycle considerations and system optimisation are closely intertwined with AI implementation, suggesting a holistic approach to system development that considers both immediate performance metrics and long-term sustainability factors. The clear yet logical connection between experimental aspects and AI-driven optimisation techniques indicates a well-balanced research field combining theoretical advancement with practical validation.

The thematic mapping of keywords in Fig. 10 visualises the research themes based on two bibliometric indicators: density and centrality. Density suggests a theme's internal strength and cohesion, indicating the extent to which the associated keywords co-occur within the research domain. In contrast, centrality indicates a theme's relevance within the broader research network and its connectivity to other thematic areas. The thematic map is divided into four quadrants: basic themes (lower right quadrant), niche themes (upper left quadrant), motor themes (upper right quadrant) and declining/emerging themes (lower left quadrant).

The basic themes quadrant contains fundamental concepts with high centrality but lower density, including solar energy, artificial neural networks, machine learning, photovoltaic, and solar water heating. These terms denote the core foundations of the field that are well-established and widely understood. Both traditional solar technologies and modern AI approaches suggest a blend of established knowledge and contemporary methods. These themes form the essential building blocks upon which more advanced research is built, and while they may not be experiencing rapid development, they remain crucial to the field's overall progress. Including genetic algorithms and renewable energy in this quadrant emphasises their fundamental importance to the field.

Likewise, the motor themes quadrant denotes well-developed and highly central themes, indicating that they are the driving forces of the research field. The motor themes comprise artificial neural networks, energy efficiency, and heat collection rate, which are crucial for optimising SWHS. The presence of cost considerations alongside technical aspects like exergy and flat-plate collector simulation demonstrates the balanced approach between technical performance and economic viability. These themes indicate the mature and well-developed areas actively advancing the field. AI-related terms in this quadrant suggest that artificial intelligence is central to improving system performance and efficiency.

Similarly, the niche themes quadrant contains highly specialised topics with high density but lower centrality. Themes in the quadrant include evacuated tube solar collectors, particle swarm optimisation, distributed energy systems, and optimal control. These themes indicate specialised research areas that, while potentially important, may not be as broadly connected to other research topics. Optimisation techniques and specific collector technologies suggest that researchers explore advanced solutions for specific challenges in solar water heating systems. These themes might represent future breakthrough areas or specialised applications that could become more central to the field as technology advances.

Furthermore, the declining theme quadrant features lower development and centrality themes, including genetic algorithms, heat pump water heaters, seasonal storage, neural networks, and CFD (computational fluid dynamics). These themes indicate established topics that, despite having been widely studied for decades and once receiving significant attention, are now attracting less research interest. Traditional technologies (heat pumps) and computational methods (CFD) suggest combining conventional approaches. These themes might denote areas that are superseded by newer technologies.

Fig. 11 displays the thematic evolution of keywords, illustrating the dynamic progression of research themes. It can be seen that the initial period (2000–2013) was primarily dominated by fundamental concepts, including artificial neural networks, solar energy, and basic optimisation techniques, establishing the groundwork for future developments.

The period from 2014 to 2018 showed a significant expansion in research themes, with the emergence of genetic algorithms as a notable optimisation tool, alongside continued development in solar energy

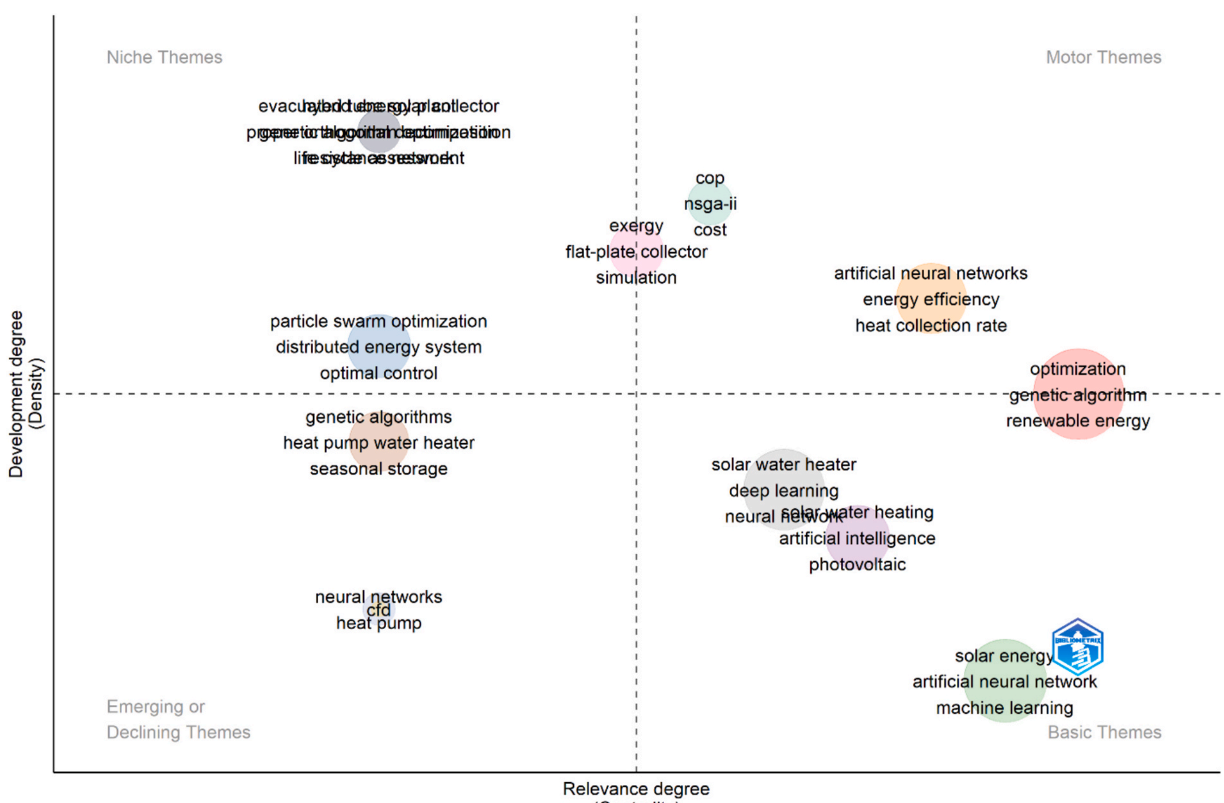


Fig. 10. Keyword-based thematic mapping.

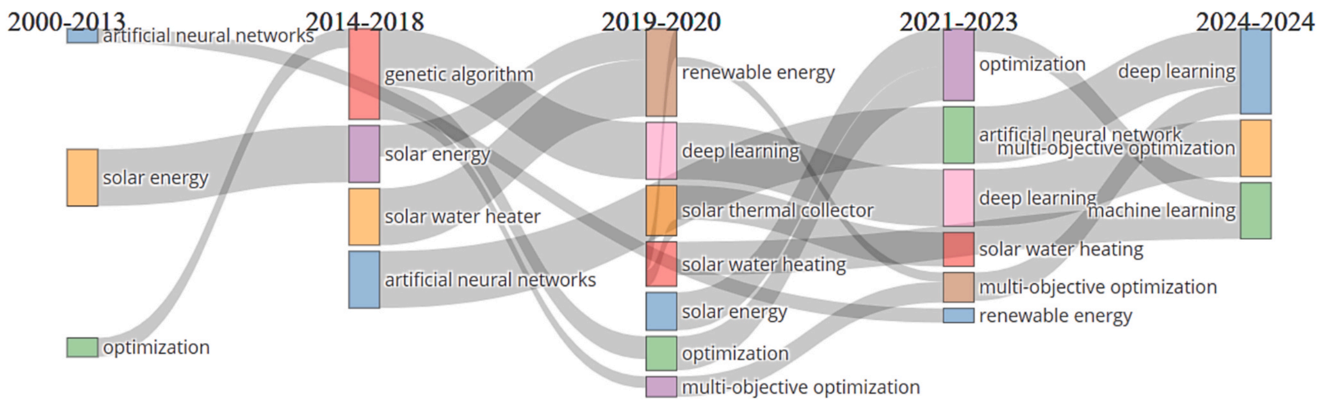


Fig. 11. Keyword-based thematic evolution analysis.

applications and solar water heater technology. The artificial neural networks research stream maintained its presence while becoming more sophisticated in its applications. During 2019–2020, the research domain witnessed a notable shift towards renewable energy integration and the introduction of deep learning techniques. Solar thermal collectors and solar water heating systems became key research focuses, emphasising practical applications. The period of 2021–2023 showed a clear evolution towards optimisation approaches, with multi-objective optimisation becoming a distinct research theme alongside continued development in artificial neural networks and deep learning applications. The latest period (2024) demonstrates the field's growth with the emergence of machine learning as a distinct theme, alongside deep learning and multi-objective optimisation. The figure shows continuity in some research streams in optimisation and neural network applications, and also reveals how newer technologies and methodologies have been incorporated.

The evolution of these themes suggests a gradual shift from more straightforward, single-objective optimisation approaches to more complex, multi-objective strategies capable of handling multiple system parameters simultaneously. The continued presence of solar energy and solar water heating themes throughout all periods indicates their fundamental importance to the field. At the same time, the emergence of newer AI techniques denotes the field's adaptation to advancing technology.

Advances in optimising SWHS with artificial intelligence

This section systematically analyses relevant studies to identify the strengths, limitations, and gaps in existing research on AI applications in SWHS. In view of this, a range of AI methodologies, from traditional approaches such as artificial neural networks and genetic algorithms to advanced deep learning architectures and hybrid intelligent systems, are discussed. Each methodology provides unique perspectives on applying computational intelligence to improve SWHS efficiency, reliability, and economic viability. The studies discussed are grouped into the following thematic areas: (1) performance prediction, modelling, and temperature forecasting using neural networks and machine learning, (2) multi-Objective optimisation, system design, and parameter sizing using evolutionary algorithms, (3) intelligent control systems, fault detection, and real-time monitoring, and (4) hybrid Systems, advanced materials integration, and specialised applications. It is worth mentioning that each thematic area indicates an individual approach to addressing specific challenges in solar water heating technology and demonstrates the breadth and depth of AI applications in this domain.

Performance prediction, modelling, and temperature forecasting using neural networks and machine learning

Performance prediction, modelling, and temperature forecasting are foundational pillars for advancing SWHS, where the integration of ANNs and machine learning techniques has revolutionised traditional analytical approaches to provide unprecedented accuracy, computational efficiency, and adaptability to complex non-linear relationships that characterise solar thermal processes. For example, ANNs have shown significant capabilities in predicting solar water heater performance across different configurations and operational conditions. However, variations in methodology, accuracy metrics, and application contexts show these computational techniques' complexity. Liu et al. [111] pioneered high-throughput screening methods using ANN models to evaluate over 350 million design combinations for water-in-glass evacuated tube solar water heaters. The authors achieved experimental validation with heat collection rates of 11.32 and 11.44 MJ/m^2 and prediction errors below 2 %. The results imply that machine learning can effectively navigate enormous design spaces that would be computationally prohibitive through conventional simulation approaches. Li & Liu [112] conducted comparative evaluations of four algorithms: general regression neural network (GRNN), multilayer feedforward neural network (MLFN), support vector machine (SVM), and extreme learning machine (ELM) and found that all achieved 100 % accuracy within ± 30 % tolerance. However, MLFN yielded the lowest RMSE of 0.14 for heat collection rate prediction. At the same time, ELM performed best for heat loss coefficients with an RMSE of 0.67, highlighting that algorithm selection depends critically on the specific performance parameter being predicted rather than the universal advantage of any single approach. This finding contrasts with Razavi et al. [30], who found that ANNs outperformed conventional heat transfer correlations with R^2 of 0.993 for training data and 0.978 for validation, compared to the conventional method's significantly reduced R^2 of 0.522 during validation. Liu et al. [28] developed a neural network that predicts solar collector thermal performance by processing eight input parameters (tube specifications, geometric configurations, and operating conditions) through two hidden layers that learn complex relationships between design variables and system behaviour, outputting the heat collection rate or heat loss coefficient to optimise collector efficiency as shown in Fig. 12. The optimal models achieved RMSE values of 0.2049 for heat collection rates and 0.9246 for heat loss coefficients. However, Casteleiro-Roca et al. [113] employed hybrid intelligent models combining ANNs and SVM across nine local models. The authors achieved predictions with less than 4 °C MAE and 14.5 % MAPE. The temporal dimension of solar thermal system behaviour has motivated investigation of recurrent architectures, with Correa-Jullian et al. [31] revealing that LSTM models achieved higher performance with MAE of 0.55 °C, RMSE of 1.27 °C, and variance of 0.520°C^2 compared to standard ANNs, RNNs, and conventional regression models, including Bayesian Ridge and Gaussian Process methods. This implies that memory-capable networks can better capture sequential dependencies and temporal patterns inherent in solar-assisted systems subject to diurnal cycles and weather fluctuations.

In the work of Sadeghi et al. [83], the authors investigated modified

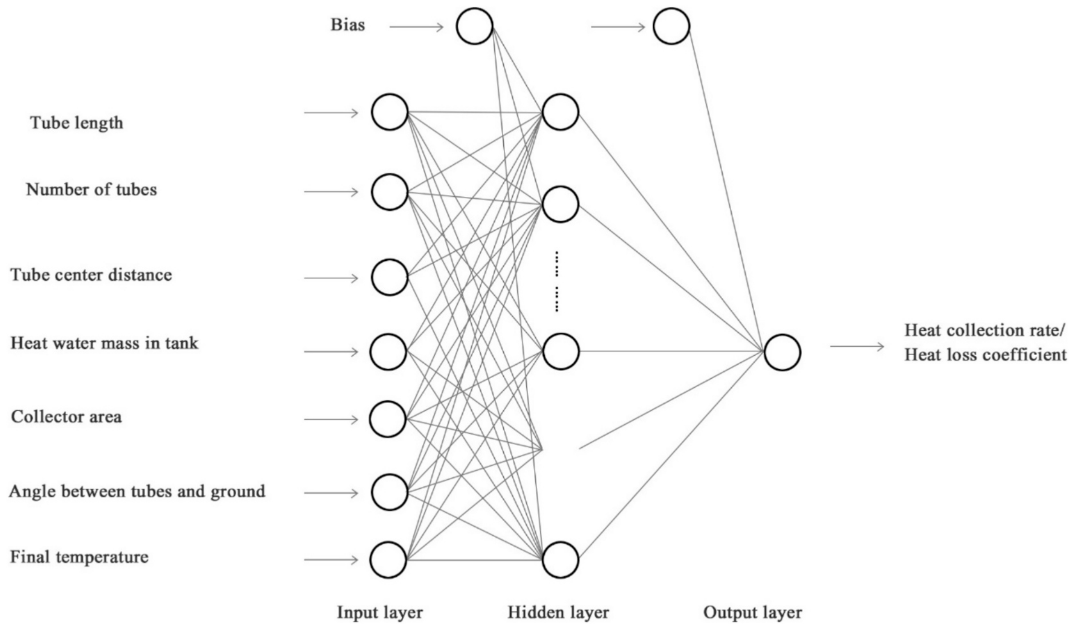


Fig. 12. Schematic ANN diagram of a neural network model for predicting SWHS heat collection rate and heat loss coefficient [28] (Published under open access).

evacuated tube collectors with nanofluid working media, comparing MLP and RBF techniques. The authors neural network model predicts thermal system performance by processing three input parameters (storage tank volume, bypass tube diameter, and nanofluid concentration) through a hidden layer to simultaneously predict two output metrics: temperature difference and energy efficiency of the thermal storage system utilizing Cu_2O/W nanofluid technology (Fig. 13). The MLP yielded better performance, recording a maximum mean relative percentage error (MRPE) of only 0.576 compared to RBF's 0.907. The choice between different collector technologies and operational conditions has influenced model performance characteristics, as evidenced by Mohana et al. [114], who analysed parabolic trough collectors and found that SVM achieved 50.99 % accuracy, substantially outperforming neural networks at 17.12 % and multivariate adaptive regression splines at 48.26 %. However, compared to simpler multiple linear regression approaches, these relatively modest absolute accuracy values raise questions about the circumstances under which complex nonlinear models provide meaningful advantages over traditional statistical methods. Prakasam et al. [84] developed an optimised ANN model with a 3–10-2 architecture, achieving an MSE of 8.2×10^{-3} and correlation coefficients ranging from 0.96 to 0.98 for flat-plate collectors using cerium oxide–water nanofluid. Comparative studies between data-driven and physics-based modelling approaches have revealed complementary strengths and weaknesses, with Shafieian et al. [115] showing that ANN is the most accurate predictor for heat pipe solar collectors with an autumn Variance Accounted For (VAF) of 0.99489 and R^2 of 0.989, significantly outperforming ANFIS and Thermal Resistance Network methods. It was also noted that prediction accuracy varied seasonally, with maximum divergences of 3.56 % for thermal efficiency and 1.52 % for exergy efficiency. Tamizharasan et al. [116] implemented deep neural networks for systems incorporating PCM, achieving training accuracy values ranging from 0.83888 to 0.98692 with RMSE values below the 15 % threshold and MAPE values mainly below 5 %. These results were substantially better than industry standards of 20–30 % MAPE and confirmed deep learning architectures' statistical significance and reliability through chi-square analysis.

Similarly, Elmasry et al. [117] developed second-order polynomial correlations using ANNs and group method data handling with a high coefficient of correlation (R^2 of 0.99824) and low RMSE (0.05109) for estimating outlet temperature. The computational efficiency advantages of machine learning have been emphasised in studies comparing training requirements and deployment complexity, with Liu et al. [118] showing that ELM models using 31 hidden nodes achieved an average RMSE of 0.30 for heat collection rates, slightly higher than previous MLFN results of 0.14 but with significantly reduced training time. In contrast, ELM with 5 hidden nodes achieved RMSE of 0.67 for heat loss coefficients, lower than all previous methods including SVM, GRNN, and MLFN. Integration with physical simulation platforms has produced hybrid modelling strategies, as Souliotis et al. [34] combined group method data handling neural networks with TRNSYS simulation for integrated collector storage systems, achieving training accuracy ($R^2 = 0.9392$) and validation ($R^2 = 0.9383$), with TRNSYS providing radiation processing capabilities. At the same time, ANNs handled black-box modelling of components that were difficult to characterise analytically. Similarly, Wongsuwan & Kumar [101] compared TRNSYS and ANN predictions for forced circulation systems. The results show that for

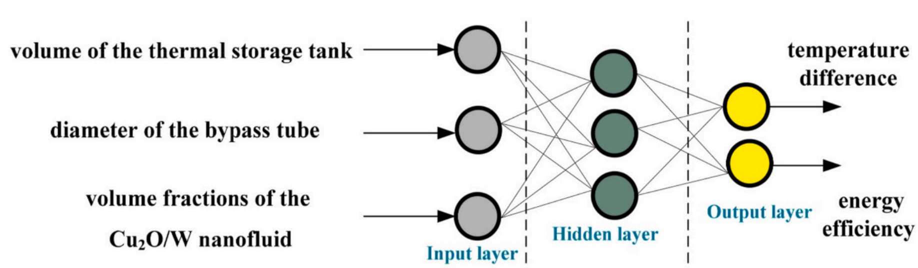


Fig. 13. ANN model for thermal storage system performance prediction (Published under open access) [83].

hourly collector outlet temperature TRNSYS achieved maximum errors of 1.5 °C MAE and 1.7 °C RMSE. In comparison, ANN showed 1.7 °C MAE and 2.1 °C RMSE. However, both achieved efficiency indices and R^2 values exceeding 0.95, with TRNSYS performing better during variable cloudiness but ANN showing higher prediction during very cloudy conditions due to sufficient training data spread.

Likewise, other studies have assessed long-term performance prediction capabilities. For instance, Kalogirou & Panteliou [29] trained ANNs on 30 thermosiphon systems and obtained an R^2 value of 0.9993 for training and 0.9913 for validation when predicting solar energy output. The results imply that networks trained on standardised test data can predict performance across collector areas ranging from 1.81 to 4.38 m^2 and multiple system configurations, including open and closed thermosiphonic designs with horizontal and vertical storage tanks. Specialised applications have motivated the development of specific predictive models, with Alade et al. [119] proposing Bayesian support vector regression for estimating specific heat capacity of alumina/ethylene glycol nanofluids, recording Pearson's correlation coefficient of 99.95 % and absolute average relative deviation of 0.1888. Ben Ali et al. [120] applied machine learning with a local regression approach, with an R^2 above 95 % for absorptance prediction in graphene-based solar absorbers. PCM integration has been successfully modelled, as Kanimozhi et al. [121] developed ANN models for thermal energy storage systems using paraffin and honey waxes. The results agree with experimental results from 265 trials with Chi-square values ranging from 1.5 to 4.8, MAPE values less than 12 %, and RMSE values between 0.65 and 1.9, with statistical analysis identifying time as the dominant factor contributing more than 40 % to heat improvement during both charging and discharging processes.

In addition, photovoltaic-integrated systems have been addressed, with Asiri et al. [122] achieving an ANN accuracy ($R^2 = 0.9998$) with less than 5 % error for predictions of solar water heaters with variable speed photovoltaic circulating pumps. Lillo-Bravo et al. [123] used random forest models to replace experimental tests required by ISO 9459–5:2007, training on 36 thermosiphon systems and testing on four systems, obtained MAPE between 2.94 % and 5.86 % and R^2 between 0.995 and 0.998 for annual energy supplied. Advanced fluid dynamics applications have emerged, with Shafiq et al. [124] investigating Darcy-Forchheimer tangent hyperbolic flow in parabolic trough collectors using the Levenberg-Marquardt technique and backpropagated neural networks. The authors created a dataset through Runge-Kutta fourth-order methods with model deviations closely clustered around zero, indicating that neural networks can capture complex rheological behaviour relevant to non-Newtonian heat transfer fluids. Thermal resistance network approaches have been enhanced through machine learning integration, as Nokhosteen & Sobhansarbandi [125] presented a resistance network-based proper orthogonal decomposition method for heat pipe evacuated tube collectors, obtaining predictions of peripheral temperature distribution with a maximum error of 10 %, combining geographical and meteorological characteristics with proper orthogonal decomposition for desired temporal accuracy.

Furthermore, storage tank design has benefited from neural network applications. Kulkarni et al. [126] used back-propagation learning with the Levenberg-Marquardt algorithm trained on 608 known values to predict the overall conductance of hot water storage tanks. A maximum error of 2.62 % was obtained, facilitating rapid and accurate generation of various possible designs considering tank diameter, insulation thickness, and thermal conductivity parameters. Thermal stratification prediction has been investigated, with Géczy-Víg & Farkas [127] comparing TRNSYS and ANN models for determining temperature distribution across eight storage tank layers and found average deviations of 1.9 °C for TRNSYS compared to 0.53 °C during training and 0.76 °C during validation for the ANN approach. In addition, fuzzy logic alternatives have been explored, with Kishor et al. [128] developing fuzzy models with three inputs achieving improved prediction performance compared to neural network techniques. Heidari and Khovalyg [33] achieved MAE reductions of 25 %, 28 %, and 41 % compared to feed-forward models for LSTM, attention-based LSTM, and attention-based LSTM with decomposition, respectively. He et al. [129] ANN models for forecasting irreversibility rates inside turbulator-inserted solar collector tubes achieved high accuracy for estimating total entropy generation rate as functions of Reynolds number and pitch distance. This allows the identification of high entropy generation regions requiring geometric modifications. Roberto et al. [130] used graybox identification and adaptive linear neural network models to infer steady-state efficiency curves from transient measurements, which recorded errors below 0.5 % and 2.2 %, respectively. Alosaimy and Hamed [131] ANN models for solar water heaters coupled with air humidifiers for liquid desiccant regeneration recorded a maximum uncertainty of 2 % in temperature measurements.

Multi-objective optimisation, system design, and parameter sizing using evolutionary algorithms

Traditional design approaches often rely on simplified heuristic methods that may fail to capture the intricate trade-offs between competing objectives such as thermal efficiency, economic viability, spatial constraints, and environmental impact. Studies in this field systematically investigate how algorithmic approaches can enhance system performance through optimal sizing of collectors, storage tanks, and auxiliary components and minimise lifecycle costs while maximising energy savings. Applying evolutionary algorithms to SWHS optimisation shows diversity in methodological approaches and optimisation objectives across different contexts and scales. Hajabdollahi and Hajabdollahi [132] employed PSO to maximise efficiency and minimise total annual cost for flat plate collectors. The study identified mass flow rate and insulator thickness as critical decision variables causing objective function conflicts, with efficiency changes of only 0.123 % to −1.506 % across different heat transfer rates but cost variations of 44.30 % to −77.87 %. Similarly, Yaman and Arslan [133] utilised a PSO/Hooke-Jeeves hybrid algorithm for optimising SWHS across Turkish regions. The authors achieved a lifecycle cost reduction of 3.3 %-5.8 % annually and 1.8 %-4.8 % during summer. Also, they revealed that optimisation parameters differ substantially between seasonal periods, with summer optimisation favouring cost savings and winter optimisation enhancing thermal performance. Ouhammou et al. [134] used a genetic algorithm to optimise active solar water heaters under varying climatic conditions, achieving a maximum thermal efficiency of 75.28 % by optimising plate emissivity, glass cover number, and air velocity parameters. Zhang et al. [135] optimised Fresnel and flat-plate collectors using NSGA-II, achieving 19 % cost reduction and 2.9 % area reduction compared to single-collector systems. Dimri and Ramousse [108] developed a genetic algorithm optimisation based on total unit product cost, revealing that a solar-assisted heat pump with photovoltaic panels achieved 41 % lower costs than reference systems. Yoshida and Ueda [136] developed a multi-objective PSO for a hybrid PV-SHWS system, which improved power grid dependence by 7.59 %, self-consumption rate by 1.41 %, and electric charge by 127.89 JPY compared to simultaneous optimisation. Novotny and Matuska [137] employed TRNSYS-MATLAB integration with NSGA-II for Czech households, revealing that electric heaters with minimal solar systems yielded the lowest costs (15 k USD). At the same time, heat pumps with extensive coverage cost the highest (43 k USD), and notably that no Pareto-optimal solution included solar thermal collectors alone due to high investment costs and low savings.

Likewise, Cao et al. [138] optimised direct-expansion solar-assisted heat pump water heaters using NSGA-II, which maximised the coefficient of performance to 95.21 % but decreased solar collector efficiency by 13.5 %. Shirinbakhsh et al. [139] reported that applying a genetic algorithm to optimise PCM integration in solar domestic hot water systems revealed that thermal stratification enhances the auxiliary solar fraction by 9 % under non-uniform high consumption conditions. In contrast, PCM integration alone increased the auxiliary solar fraction by

4 %, though the improvement was highly case-dependent, with demand variations causing up to an 8 % change in the auxiliary solar fraction. Silva et al. [95] employed CFD and the NSGA-II optimisation algorithm to optimise stamped longitudinal vortex generators in circular tubes. Their results showed that heat transfer was enhanced by more than 50 % compared to smooth tubes, with the highest improvement of 62 % at a Reynolds number of 900. However, the optimal thermo-hydraulic efficiency was achieved at a Reynolds number of 600, where the heat transfer gain accounted for 55 % of the corresponding pressure drop penalty. Mayer et al. [140] applied a multi-objective genetic algorithm to optimise waffle-shaped flat-plate collectors. The optimal configuration achieved a solar absorptance of 97.8 % and a thermal emittance of 4.8 %, representing significant improvements compared to flat configurations, which showed 84.9 % absorptance and 1.7 % emittance. Atia et al. [141] employed a genetic algorithm to optimise the solar collector area, achieving an annual solar fraction of 98 %. Seasonal variations were observed, with solar fractions ranging from 90-99 % in summer and 44.9–-2.8 % in winter. In comparison, economic analysis revealed system costs increased 157 % when interest rates increased from 5 % to 20 %. Ponnusamy et al. [90] performed an exergetic characterisation using a genetic algorithm to minimise the overall loss coefficient through optimisation of heat flux rate and glass cover temperatures, finding that exergy efficiency increased slightly (0.01–0.08 %) over 12 h, with the minimum loss coefficient occurring during peak hours. Ahmadi et al. [96] NSGA-II optimisation distillation and compact solar water heaters identified optimal parameters of 20 membrane rows, 334.4 kg feed mass, and 0.027 kg/s flow rate, achieving 92.5 kg daily production at 12.2 USD/m^3 cost with gain output ratio of 1.6.

Similarly, Fong et al. [142] applied evolutionary programming with 10 individuals and a maximum of 50 epochs, achieving annual energy savings of 1561.9 GJ at optimal conditions comprising a 20.6° tilt angle, 15.5° southwest orientation, 41.6 m^3 storage capacity, and a pump flow rate of 20,427 kg/h. Adun et al. [143] employed a genetic algorithm to optimise ternary nanofluids in parabolic trough collectors integrated with Kalina cycles, attaining a maximum power output of 61.21 kW and an optimum exergetic efficiency of 25.1 %, outperforming conventional thermal oils and salts. Kumar et al. [144] experimentally demonstrated that angled perforated baffles increased the Nusselt number and friction factor by factors of 3.82 and 7.14, respectively, achieving optimal performance at a 55° attack angle and a Reynolds number of 9,000, with a thermohydraulic performance of 1.94. The central composite design and ANN models accurately predicted the thermal parameters, with deviations within ± 9.7 % and ± 5.0 %, respectively. Khargotra et al. [145] identified the optimal configuration with a 0.5 pitch ratio, 0.2 blockage ratio, 45° attack angle, and 0 mm spacer length, where empirical correlations and ANN predictions showed bias errors of 10.9 % for the Nusselt number and 9.5 % for the friction factor.

Intelligent control systems, fault detection, and real-time monitoring

This section highlights studies integrating intelligent control systems, fault detection, and real-time monitoring in SWHS. For example, Cardona et al. [146] developed a wireless IoT-based temperature monitoring system utilising machine learning algorithms for intelligent alarm triggering, including water leak detection and efficiency loss identification, delivering real-time data through local Wi-Fi networks for user notification and automation integration. Chidiac et al. [87] employed an adaptive self-organising map neural network that achieved convergence after 200 training iterations using 2,208 samples. The optimal approach, utilising adaptive shading of 0 %, 20 %, 25 %, or 32 %, maintained collector temperatures within 2.05 °C of the 80 °C limit and prevented unacceptable nighttime temperature drops observed with more aggressive shading strategies. Akbari and Shadlu [147] addressed temperature profile control challenges through fuzzy logic-modified model predictive control. The speed and accuracy recorded were higher than those of traditional model predictive control in MATLAB/Simulink simulations, with the fuzzy logic component calculating state variable references while modified model predictive control regulated temperature parameters despite system disturbances and delays. The model-free paradigm advances further in Pataro et al. [148], where reinforcement Q-Learning achieved adaptive optimal control of solar collector fields using only plant measurements (working fluid temperature, flow rate, and reference trajectory). The algorithm converged to optimal linear quadratic tracking (LQT) controller gains, showing less than 15 % average discrepancy across 18,000 data points while outperforming LQT in managing steady-state errors during smooth solar irradiance conditions. Ahmed et al. [149] optimised PV/T hybrid collectors using fuzzy logic monitoring reflective mirrors, glass covers, and lower reflector angles as inputs with PV temperature and efficiency as outputs, achieving thermal efficiency variations from 82 % to 50 % depending on angle configurations.

Likewise, Nguyen et al. [150] employed a PID temperature controller for passive systems to obtain less than 3 % overshoot, approximately 110-second settling time, and ± 1°C temperature error and maintained desired outlet temperatures through hot–cold water mixing ratio adjustments. Atia et al. [151] compared neural network control against adaptive fuzzy logic control (AFLC) for aquaculture applications trained via Levenberg-Marquardt back-propagation and obtained a perfect R = 1.0 regression and maintained exactly 34 °C throughout 24-hour periods in both seasons. In contrast, the AFLC recorded an initial 2 °C overshoot, potentially harmful to aquaculture organisms. The comparative analysis of control strategies extends to Araújo et al. [152], who revealed that proportional flow rate control outperforms on–off control across medium-to-high collector areas, with proportional control achieving maximum solar fractions exceeding on–off control by factors of 1.45–1.78 at a standard 5 m^2 collector area, depending on location and collector type. However, on–off control proved marginally higher only for impractically small areas (A $\lesssim$ 0.15 m^2) where solar fractions remained extremely low at 0.05–0.07.

Other studies have assessed fault detection capabilities utilising different machine learning architectures and sensor fusion strategies. For example, Jiang et al. [153] achieved 89.7 % minimum accuracy with a 93.7 % mean across 100 tests using SVM-Dempster-Shafer fusion to detect four fault types (dusty collectors, heating empty collectors, heat transfer inefficiency, and leakage). The Dempster-Shafer evidence theory fusion yielded fused basic belief values of 0.9234–0.9630 with uncertainties below 1 %, representing improvements over individual evidence sources, where audio features alone showed 0.6533 confidence and temperature evidence only 0.4378 for dusty collector faults. Kalogirou et al. [154] developed eight ANNs (Fig. 14) for fault detection, achieving R^2 values of 0.8258–0.9996 for validation datasets, with the M_{10}a method proving more stable than M_5c due to ten-consecutive-measurement averaging. The models detected collector efficiency degradation at approximately 5 % loss (F_0 coefficient declining from 0.70 to 0.67) at the 76th-80th functioning hours and insulation deterioration at 5 % degradation levels at the 67th-99th hours, depending on the detection method and circuit location. He et al. [155] implemented

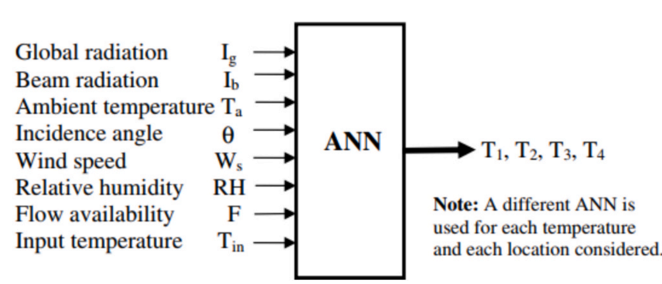


Fig. 14. A schematic of an ANN model for SWHS fault predicting temperatures [154] (Reproduced with permission from Elsevier, License number: 5924950460908).

hierarchical ART networks trained on five-year TRNSYS January simulations using only two sensors (collector plate and tank heat exchanger outlet temperature). With vigilance parameters ranging from 0.65 to 0.87, creating 3–14 pattern categories across four hierarchical layers, the model detected pump failure rapidly, pump degradation progressively at 40 % flow reduction, thermosyphoning at low-to-medium severity during nighttime reverse circulation, and persistent shading requiring 5-day mean and standard deviation inputs to distinguish from transient cloud coverage. The application of advanced computer vision techniques emerges in Tao et al. [156], who developed TMACNet based on modified YOLOv8 Siamese architecture with feature difference branch, temporal mutual attention layer, and contextual attention module for small object change detection in UAV imagery of rooftop solar water heaters. TMACNet achieved F1 improvements exceeding 5.96 % compared to alternative change detection methods and 1–3 % AP metric improvements over single-temporal detection models, and shows resistance to image registration errors and building height displacement.

Hybrid Systems, advanced materials integration, and specialised applications

Studies discussed in this section highlight hybrid systems combining multiple solar technologies, such as PV/T collectors, absorption heat transformers, or PCM integration, to address the inherent limitations, such as intermittency challenges, temperature elevation constraints, and spatial footprint requirements. Integrating PCM have substantial performance enhancements across different collector configurations and operational settings in this context. For example, Pathak et al. [157] evaluated annular copper fin integration with U-tube evacuated tube collectors incorporating PCMs. The annular copper fin U-tube evacuated tube collector with PCM achieved 73.85–87.84 % for simultaneous charging mode and 71.23 85.44 % for midday charging mode thermal output, enhancing performance by 9–21 % over the reference system and reducing heat loss. Chopra et al. [158] integrated myristic acid within novel manifold designs for evacuated tube solar collectors. The authors attained daily efficiencies of 62.42 %-79.12 % during half-day charging and 71.49 %-85.21 % during whole-day charging, depending on flow rates (0.066–0.20 L/min), representing 42.85 % and 61 % improvements over conventional systems, respectively, with higher flow rates recommended for maximising daily efficiency. In comparison, lower rates produced higher outlet temperatures suitable for late evening demand.

The hybrid system paradigm advances significantly in López-Pérez et al. [159], who evaluated SWHS coupled with absorption heat transformers (AHT) across three Mexican climates. The results show that SWHS-AHT configurations attained collector area reductions of 42.9 %-60 % and sustained comparable auxiliary heat requirements, with the most notable results in Tapachula requiring only 150 m^2 versus 375 m^2 for conventional systems (60 % reduction) and generating potential investment savings of €86,512.5. At the same time, the AHT maintained a COP of 0.44 and achieved an average temperature lift of 12.9 °C across all locations. In the work of [160], the authors proposed a hybrid solar deep learning approach incorporating automatic sun-tracking and defective tube lamps with methane gas heat conduction. The model claims 86–92 % performance rates across input sizes and management categories, representing improvements of 10–20 % points over traditional building-integrated photovoltaic systems. Deepanraj et al. [32] modified shallow swarm optimisation with deep learning employing restricted Boltzmann machine models with quasi-oppositional-based learning for hyperparameter optimisation, achieving prediction accuracy with MSE of 0.0066, RMSE of 0.0811, and MAE of 0.0616. The computational modelling sophistication reaches new heights in Nokhosteen and Sobhansarbandi [161], who developed an innovative hybrid CFD model combining resistance network-based proper orthogonal decomposition for daytime operation with the Lattice Boltzmann method for nighttime simulation. The model agreed well with experimental data (average error 5.16 %) while operating several orders of magnitude faster than traditional direct numerical simulation methods, addressing the computational intensity challenge inherent in modelling complex heat transfer processes within evacuated tube collectors across full operational cycles. Shaban et al. [162] employed optimised deep neural network (ODNN) enhanced with the sand cat swarm optimisation (SCSO) algorithm to determine the optimal design of PV/T cooling systems. The findings revealed that the ODNN model achieved high predictive accuracy with MAE, MSE, and R^2 values of approximately 0.0075, 0.00251, and 0.9972, respectively.

Summary and future research directions

This section summarises the bibliometric and systematic review findings and identifies potential directions for future research. The bibliometric analysis revealed that China leads with 22 % of the total research output, followed by India with 16 % and the United States with 10 %. In contrast, Africa contributes only 5.3 %, emphasising a significant knowledge gap despite the continent's vast solar energy potential. ANNs achieved prediction accuracies with R^2 values exceeding 0.99 for performance forecasting across multiple system configurations. Table 1 summarises AI methodologies applied to SWHS for performance prediction, fault detection, and intelligent control, including their parameters, performance metrics, and key outcomes. Advanced architectures, including LSTM networks, revealed higher temporal prediction capabilities with MAE as low as 0.55 °C [31]. Evolutionary algorithms have addressed multi-objective optimisation challenges, balancing thermal efficiency, lifecycle costs, and environmental impacts. Genetic algorithms and PSO identified Pareto-optimal designs that reduced lifecycle costs by 3–20 % and maintained performance [92,108,133]. Intelligent control systems incorporating fuzzy logic, reinforcement learning, and neural networks supported adaptive operation that outperformed conventional controllers [147,148,151]. Fault detection systems achieved accuracies exceeding 89 % using sensor fusion and machine learning [153,154]. PCM integration enhanced energy storage by 5–62 % depending on materials and configurations [140,141,158]. Hybrid photovoltaic-thermal systems and absorption heat transformer coupling reduced collector areas by 43–60 % and maintained thermal performance [159].

Despite these advances, critical limitations persist and require further investigation. The transferability of trained models across geographical locations and climatic zones remains insufficiently explored, as most studies validated algorithms using site-specific data [29,83,101]. Future research could develop generalised models trained on multi-location datasets representing several climate classifications and operational conditions similar to approaches that achieved high accuracy across varying environmental parameters [70,123]. Long-term model degradation and recalibration requirements warrant systematic investigation since solar systems operate for 15–25 years under changing environmental conditions. Further studies could establish maintenance protocols for retraining algorithms as system performance evolves, given demonstrated capability to detect efficiency degradation at approximately 5 % loss levels [154]. Uncertainty quantification is another critical gap, as most studies reported deterministic predictions without confidence intervals or probabilistic forecasts [111,112,118]. Future work should integrate Bayesian approaches or ensemble methods, providing prediction uncertainty estimates essential for risk-informed decision-making for applications requiring precise temperature control [119,151].

Likewise, the computational intensity of deep learning architectures limits deployment on low-cost embedded systems common in solar installations [31,116]. Further research could focus on model compression techniques, edge computing implementations, and hardware-aware neural architecture search, enabling real-time optimisation on resource-constrained platforms, addressing challenges where advanced controllers demonstrated better performance but required substantial

Table 1
Summary of AI techniques for SWHS performance prediction, fault detection, and intelligent control.

AI methods used	Key parameters/variables	Performance metrics	Main outcomes	Reference
ANN, RNN, LSTM	Meteorological conditions, system performance	MAE = 0.55 °C, RMSE = 1.27 °C, Variance = 0.520°C^2, Relative error = 3.45 %	LSTM models outperformed ANN and RNN with the lowest prediction errors for temperature sequences; Temporal behaviour replication	[31]
GRNN, MLFN, SVM, ELM	Heat collection rate, heat loss coefficient	100 % accuracy at ± 30 % tolerance; RMS: 0.14–0.33 (heat collection), 0.67–0.73 (heat loss)	MLFN is best for heat collection (0.14), ELM is best for heat loss (0.67)	[112]
ANN-based high-throughput screening	Extrinsic properties of water-in-glass evacuated tube solar water heater (3.538×10^8 combinations)	Prediction error: 1.35–1.90 %, Heat collection: 11.32–11.44 MJ/m^2	Screened millions of designs efficiently; two new designs showed higher rates than all 915 measured samples	[111]
Attention-based LSTM + Time-series decomposition (ALSTM-D)	Demand pattern, solar radiation (stochastic phenomena)	MAE reduction: 41 % vs Feed-Forward, 28 % vs LSTM, 25 % vs ALSTM	Novel attention mechanism for solar-assisted systems; Higher handling of stochastic inputs	[33]
MLP and RBF	Tank volume (5–8 L), bypass pipe diameter (6–10 mm), nanofluid concentration (0–0.04)	MLP: MRPE = 0.576 %, RBF: MRPE = 0.907 %; Energy efficiency improvement: 4 %	MLP higher than RBF; validated with a modified evacuated tube collector using Cu_2O/DW nanofluid	[83]
Modified shallow swarm optimisation + Deep Learning (MSSODL-EUP) with restricted Boltzmann machine (RBM)	Energy use prediction using RBM	MSE = 0.0066, RMSE = 0.0811, MAE = 0.0616	The MSSODL technique with QOBL optimisation outperformed competing algorithms in energy utilisation prediction.	[32]
Back-propagation neural network	Tube length, number, diameter, tank volume, angle, final temperature	RMSE = 0.2049 (heat collection), 0.9246 (heat loss); Prediction error < 0.5 %	Reduced testing time from 15 days to real-time; 915 samples analysed; computers and Android platforms	[28]
Feed-forward ANN (6–6-1 configuration)	Tube material, flow rate, temperature, and solar radiation	R^2 = 0.993 (training), 0.978 (validation) vs 0.808–0.522 (conventional)	ANN outperformed conventional heat transfer correlations by 18.5–45.6 % points	[30]
ANN (3–10-2 architecture)	Flow rate (1–3 L/min), CeO_2/water nanofluid (0.01 % concentration)	MSE = 8.2×10^{-3}, Correlation = 0.96–0.98, Max efficiency = 78.2 % at 2 L/min	21.5 % thermal efficiency increase vs plain water; holistic approach combining analytical and field testing	[84]
Deep neural network (DNN)	Inlet/outlet temps, ambient temp, tank temps, hot water flow, electricity demand	Training accuracy: 0.839–0.987, Testing: 0.720–0.987; RMSE: 0.0131–0.3126, MAPE < 15 %	DNN model for SWHS with PCM; all error metrics below 15 % threshold	[116]
ANNs (2 networks)	Collector area (1.81–4.38 m^2), system characteristics, climatic data, and demand temperature	R^2 = 0.9993 (network 1), 0.9848–0.9926 (network 2); Validation: 0.9913–0.9940	30 thermosiphon domestic SWHS tested per ISO 9459–2; pioneering long-term performance prediction	[29]
ANN, ANFIS, thermal resistance network	Seasonal variations, thermal/exergy efficiency	ANN: R^2 = 0.989 (autumn), VAF = 0.99489; Max divergence: 3.56 %	Comprehensive comparison across 4 seasons; ANN is higher based on the R^2 parameter; heat pipe solar collectors	[163]
Hybrid: ANN + SVMs (9 local models)	Solar thermal collector output temperature for smartgrid/ buildings	MAE < 4°C, MAPE < 14.5 %, MSE = 30.5010, NMSE = 0.1144	Hybrid configuration with clustering approach; more accurate than global models	[113]
TRNSYS + ANN (comparative study)	Dual storage tanks (200 + 216 L), collector outlet temp, useful energy	TRNSYS: MAE = 1.5 °C, ANN: MAE = 1.7 °C; Both R^2 > 0.95	14-day tropical climate study; ANN better for very cloudy days, TRNSYS better for partly cloudy	[101]
SVM-DS fusion (SVM + Dempster-Shafer)	Audio features (8-dim), temperature (5-dim), flow (5-dim), heat transfer rate (5-dim)	Accuracy: 89.7–93.7 % (proposed) vs 77.6–84.7 % (traditional); Confidence > 90 %, Uncertainty < 1 %	Multi-feature fusion for soft fault detection; 4 fault types detected; PSO optimisation (c = 6.673, g = 0.154)	[153]
ANNs (8 networks)	Collector efficiency degradation, pipe insulation deterioration	R^2 = 0.8258–0.9996; Detection at ~ 5 % efficiency loss	Fault diagnostic system using TRNSYS; M_{10}a method is more stable than M_5c; early alarm capability	[154]
ELM	Heat collection rate, heat loss coefficient (915 samples)	RMSE = 0.30 (heat collection), 0.67 (heat loss); 85 % training, 15 % testing	ELM provided lower RMSE for heat loss coefficients than SVM, GRNN, and MLFN, making it practical for industrial applications	[118]
ANN with Levenberg-Marquardt (608 known values)	Tank diameter, insulation thickness, conductivity, and Overall Conductance	Max error = 2.62 %	Innovative design approach for hot water storage tank; 612 design combinations analysed	[126]
ANN + group model data handling	Reynolds number (800–2000), twisted pitch distance ξ (100–200 mm)	R^2 = 0.99824, RMSE = 0.05109	Second-order polynomial with six coefficients for outlet temperature prediction; double-twisted tape turbulator	[117]
Neural network control (NNC) vs Adaptive fuzzy logic control (AFLC)	Air temperature, pond temperature, error signal, and aquaculture application	NNC: R = 1.0, perfect 34 °C control; AFLC: 36 °C overshoot	NNC superior with 3-input/7-hidden/1-output architecture; Levenberg-Marquardt training; 24-hour control	[151]
Random forest	System characteristics, climatic data, and daily load volume	MAPE: 2.94–5.86 %, MAE: varies, R^2: 0.995–0.998 (energy), 0.973–0.976 (solar fraction)	Random forest model replaced ISO 9459–5 experimental tests; 36 thermosiphon SWHS training, 4 testing	[123]
ART neural networks (hierarchical)	Collector plate temp, tank heat exchanger outlet temp (only 2 sensors)	4 severity layers; vigilance parameters: 0.65–0.87	Detected 4 fault types (pump failure, degradation, thermosyphoning, shading); TRNSYS training (2000–2004)	[155]
Graybox state space model + Dynamic adaptive linear neural network (ALNN)	Transient measurements (solar radiation, mass flow, temperature)	Graybox: <0.5 % error, ALNN: <2.2 % error vs certified values; 94 % correspondence (Graybox)	Inferred steady-state efficiency from 4 days of transient data; evacuated tube Dewar pipe collector	[130]

(continued on next page)

Table 1 (*continued*)

AI methods used	Key parameters/variables	Performance metrics	Main outcomes	Reference
ANN	PV array voltage, radiation intensity, and flow output for variable speed pump	$R^2 = 0.9998$, error < 5 % for vast majority	ANN sensitivity modelling for PV arrays; hydraulic/thermal modelling with Crank-Nicolson method	[122]
TRNSYS + ANN (8-layer model)	8-layer thermal stratification, solar radiation, water consumption, ambient temp, mass flow rate	TRNSYS deviation: 1.9 °C, ANN training: 0.53 °C, validation: 0.76 °C	ANN model (8 tansig + 8 linear neurons) outperformed TRNSYS in predicting thermal stratification	[127]

computational resources [148]. Physics-informed neural networks combining first-principles knowledge with data-driven learning remain underexplored despite potential advantages for extrapolation beyond training data ranges [34,101]. Future investigations could develop hybrid modelling frameworks incorporating thermodynamic constraints, energy balance equations, and material properties within neural network architectures, building on the demonstrated success of TRNSYS-neural network integration that achieved R^2 exceeding 0.93 [34,102]. The training data requirements and optimal sampling strategies lack systematic characterisation, with studies employing datasets ranging from hundreds to hundreds of thousands of samples [28,87,111]. Further studies should establish minimum data requirements across different algorithms, system types, and prediction objectives and develop active learning strategies that minimise experimental testing burdens, relevant given demonstrated capability to reduce conventional testing time from 15 days to instantaneous predictions [28,123].

Similarly, Africa's lower contribution to this research domain demands urgent capacity-building initiatives and contextualised research programmes. Future studies should investigate system adaptations for off-grid rural communities, integration with existing energy infrastructure, and cost-reduction strategies suitable for developing economies, addressing the critical risk that AI-driven solutions will be designed without consideration for African contexts, climatic conditions, and economic constraints. Multi-objective optimisation studies focused on thermal efficiency and lifecycle costs while neglecting environmental impact indicators, embodied energy, and social acceptance factors [92,132,138,[58,73,92,132,138]. Future research should incorporate comprehensive sustainability metrics, including carbon footprint, water consumption, and circular economy considerations extending beyond approaches that achieved substantial primary energy savings but did not fully quantify environmental benefits [106,109]. Practical validation of intelligent control systems remains limited, as most demonstrations occurred in simulation environments or short-duration experiments [147,148,152]. Long-term field studies across different climates, building types, and user behaviours should validate performance claims and identify practical implementation challenges, notably since demonstrated laboratory achievements, including 12–22 % electricity savings, require validation under realistic operating conditions [88,146]. In addition, integrating smart grid infrastructure, demand response programmes, and multi-energy systems is an emerging opportunity inadequately addressed in current literature [75,110,136]. Future research could explore predictive control strategies leveraging weather forecasts, dynamic electricity pricing, and building thermal models for holistic energy management, building on successful model predictive control demonstrations that balanced multiple objectives simultaneously [104,147,148]. These coordinated research directions would accelerate solar thermal technology deployment and ensure equitable global access to sustainable water heating solutions.

Conclusion

This study systematically assessed the integration of AI methodologies into SWHS through a dual framework combining bibliometric and systematic review analyses. The findings indicate a fast-expanding research area that has evolved from early experimental approaches to advanced data-driven optimisation and intelligent control strategies. The publication trends indicate a growth in research activity, after 2019, driven by the global interest in renewable energy technologies and AI's proven capability to enhance system performance. China, India, and the United States account for nearly half of the global research output. In contrast, Africa's marginal contribution of only 5.3 % highlights a critical knowledge gap that must be addressed to ensure equitable technological diffusion and local adaptation of AI-enhanced SWHS across solar-rich but energy-poor regions. The thematic and keyword analyses revealed five main research clusters comprising performance prediction using neural networks and machine learning, multi-objective optimisation through evolutionary algorithms, intelligent control and fault detection, neural network-simulation hybridisation using TRNSYS, and the emerging application of deep learning and hybrid photovoltaic-thermal systems. These clusters consistently focus on improving energy efficiency, thermal performance, and cost-effectiveness, exploring new frontiers in predictive maintenance and real-time adaptive control. Nevertheless, the review identified significant research gaps, including limited experimental validation of AI models, a lack of standardised datasets, minimal integration of hybrid physics-informed AI models, and weak interdisciplinary collaboration between computer science and solar engineering communities. In addition, socio-economic and policy dimensions remain scarcely explored, particularly in cost-benefit analysis, scalability, and adoption in developing regions. The study's bibliometric insights show an uneven global research network dominated by Northern Hemisphere countries, with limited South-South collaborations. This imbalance highlights the urgent need to strengthen international partnerships and capacity building in regions with limited research visibility, especially Africa and parts of Latin America. AI-enabled SWHS requires technological innovation, institutional support, data-sharing frameworks, and inclusive research funding mechanisms. This review establishes that AI transforms SWHS from static, manually controlled devices into dynamic, intelligent, and self-optimising systems capable of delivering higher efficiency, reliability, and sustainability. AI facilitates smarter utilisation of solar resources, reduces operational costs, and extends system lifespan by integrating predictive analytics, optimisation algorithms, and intelligent control. Future research should prioritise hybrid AI-physics models, real-world validation, and socio-technical applications to ensure that these intelligent systems contribute meaningfully to global energy decarbonisation and sustainable development goals.

CRediT authorship contribution statement

Flavio Odoi-Yorke: Writing – review & editing, Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Validation, Visualization.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper..

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Data availability

No data was used for the research described in the article.

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